



basic education

Department:
Basic Education
REPUBLIC OF SOUTH AFRICA

Curriculum and Assessment Policy Statement

WELDING

YEAR 1-4

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SECTION 1:

INTRODUCTION TO THE CURRICULUM AND ASSESSMENT POLICY STATEMENT

1.1 BACKGROUND

The *National Curriculum Statement Grades R-12 (NCS)* stipulates policy on curriculum and assessment in the schooling sector.

To improve implementation, the National Curriculum Statement was amended, with the amendments coming into effect in January 2012. A single comprehensive Curriculum and Assessment Policy document was developed for each subject to replace Subject statements, Learning Programme Guidelines and Subject Assessment Guidelines in Grades R-12.

1.2 OVERVIEW

- (a) The *National Curriculum Statement Grades R-12 (January 2012)* represents a policy statement for learning and teaching in South African schools and comprises the following:
- (i) *Curriculum and Assessment Policy Statements for each approved school subject;*
 - (ii) *The policy document, National policy pertaining to the programme and promotion requirements of the National Curriculum Statement Grades R-12; and*
 - (iii) *The policy document, National Protocol for Assessment Grades R-12 (January 2012).*
- (b) The *National Curriculum Statement Grades R-12 (January 2012)* replaces the two current national curricula statements, namely the
- (i) *Revised National Curriculum Statement Grades R-9, Government Gazette No. 23406 of 31 May 2002, and*
 - (ii) *National Curriculum Statement Grades 10-12 Government Gazettes, No. 25545 of 6 October 2003 and No. 27594 of 17 May 2005.*
- (c) The national curriculum statements contemplated in subparagraphs b(i) and (ii) comprise the following policy documents which will be incrementally repealed by the *National Curriculum Statement Grades R-12 (January 2012)* during the period 2012-2014:
- (i) *The Learning Area/Subject Statements, Learning Programme Guidelines and Subject Assessment Guidelines for Grades R-9 and Grades 10-12;*
 - (ii) *The policy document, National Policy on assessment and qualifications for schools in the General Education and Training Band, promulgated in Government Notice No. 124 in Government Gazette No. 29626 of 12 February 2007;*

- (iii) *The policy document, the National Senior Certificate: A qualification at Level 4 on the National Qualifications Framework (NQF), promulgated in Government Gazette No.27819 of 20 July 2005;*
 - (iv) *The policy document, An addendum to the policy document, the National Senior Certificate: A qualification at Level 4 on the National Qualifications Framework (NQF), regarding learners with special needs, published in Government Gazette, No.29466 of 11 December 2006, is incorporated in the policy document, National policy pertaining to the programme and promotion requirements of the National Curriculum Statement Grades R-12; and*
 - (v) *The policy document, An addendum to the policy document, the National Senior Certificate: A qualification at Level 4 on the National Qualifications Framework (NQF), regarding the National Protocol for Assessment (Grades R-12), promulgated in Government Notice No.1267 in Government Gazette No. 29467 of 11 December 2006.*
- (d) The policy document, *National policy pertaining to the programme and promotion requirements of the National Curriculum Statement Grades R-12*, and the sections on the Curriculum and Assessment Policy as contemplated in Chapters 2, 3 and 4 of this document constitute the norms and standards of the *National Curriculum Statement Grades R-12*. It will therefore, in terms of section 6A of the *South African Schools Act, 1996 (Act No. 84 of 1996)*, form the basis for the Minister of Basic Education to determine minimum outcomes and standards, as well as the processes and procedures for the assessment of learner achievement to be applicable to public and independent schools.

1.3 GENERAL AIMS OF THE SOUTH AFRICAN CURRICULUM

- (a) The *National Curriculum Statement Grades R-12* gives expression to the knowledge, skills and values worth learning in South African schools. This curriculum aims to ensure that children acquire and apply knowledge and skills in ways that are meaningful to their own lives. In this regard, the curriculum promotes knowledge in local contexts, while being sensitive to global imperatives.
- (b) The *National Curriculum Statement Grades R-12* serves the purposes of:
 - equipping learners, irrespective of their socio-economic background, race, gender, physical ability or intellectual ability, with the knowledge, skills and values necessary for self-fulfilment, and meaningful participation in society as citizens of a free country;
 - providing access to higher education;
 - facilitating the transition of learners from education institutions to the workplace; and
 - providing employers with a sufficient profile of a learner's competences.
- (c) The *National Curriculum Statement Grades R-12* is based on the following principles:
 - Social transformation: ensuring that the educational imbalances of the past are redressed, and that equal educational opportunities are provided for all sections of the population;

- Active and critical learning: encouraging an active and critical approach to learning, rather than rote and uncritical learning of given truths;
- High knowledge and high skills: the minimum standards of knowledge and skills to be achieved at each grade are specified and set high, achievable standards in all subjects;
- Progression: content and context of each grade shows progression from simple to complex;
- Human rights, inclusivity, environmental and social justice: infusing the principles and practices of social and environmental justice and human rights as defined in the Constitution of the Republic of South Africa. The National Curriculum Statement Grades R-12 is sensitive to issues of diversity such as poverty, inequality, race, gender, language, age, disability and other factors;
- Valuing indigenous knowledge systems: acknowledging the rich history and heritage of this country as important contributors to nurturing the values contained in the Constitution; and
- Credibility, quality and efficiency: providing an education that is comparable in quality, breadth and depth to those of other countries.

(d) The National Curriculum Statement Grades R-12 aims to produce learners that are able to:

- identify and solve problems and make decisions using critical and creative thinking;
- work effectively as individuals and with others as members of a team;
- organise and manage themselves and their activities responsibly and effectively;
- collect, analyse, organise and critically evaluate information;
- communicate effectively using visual, symbolic and/or language skills in various modes;
- use science and technology effectively and critically showing responsibility towards the environment and the health of others; and
- demonstrate an understanding of the world as a set of related systems by recognising that problem solving contexts do not exist in isolation.

(e) Inclusivity should become a central part of the organisation, planning and teaching at each school. This can only happen if all teachers have a sound understanding of how to recognise and address barriers to learning, and how to plan for diversity.

The key to managing inclusivity is ensuring that barriers are identified and addressed by all the relevant support structures within the school community, including teachers, District-Based Support Teams, Institutional-Level Support Teams, parents and Special Schools as Resource Centres. To address barriers in the classroom, teachers should use various curriculum differentiation strategies such as those included in the Department of Basic Education's *Guidelines for Inclusive Teaching and Learning* (2010).

To cater for inclusivity, the Curriculum and Assessment Policy Statement has been expanded to include learners participating in the General Education and Training (GET) Band who have an interest and talent in applied knowledge in the technical, and vocational and occupational skills. The various subjects are introduced to give learners an opportunity to achieve a formal qualification or recognition of achievement towards a qualification that is related to technical and vocational, and occupational learning within their area of interest and talents.

The Curriculum and Assessment Policy Statement: Year 1- 4 responds more effectively to the needs of learners who have been identified and assessed through the protocols approved by the Department of Basic Education. These learners will benefit from curriculum content that is aligned to Grades 8 and 9 of the General Education and Training Band at a more applied and functional level in accordance with their interest and talents.

1.4 TIME ALLOCATION

1.4.1 Year 1 - 4

(a) The instructional time in the Year 1 - 4 is as follows:

Instructional Time for the Year 1 – 4 is 27½ hours in a five-day cycle

Subjects		Time
Fundamental Subjects		
Languages (Home Language and First Additional Language)		3 Hours for Home Language
All 11 official languages (Afrikaans, English, isiNdebele, isiXhosa, isiZulu, Siswati, Sesotho, Setswana, Sepedi, Tshivenda, Xitsonga)		2 hours for First Additional Language
Mathematics		3 hours
Life Skills	Areas of Study	
	Personal and Social Well-being (including aspects of Life Orientation, Social Sciences and Economic and Management Sciences)	2½ hours
	Physical Education	1 hour
	Creative Arts	1 hour
	Natural Sciences	1½ hours from year 2 onwards This time to be used in year 1 to support
		6 hours

		Languages and Mathematics	
Information Communication Technology ICT is a compulsory subject for all learners. It can be offered either as a stand-alone or integrated across various subjects. If offered as a stand-alone a school may use time allocated to the Occupational programme. ICT does not count towards the qualification but is a necessary life-long skill. ICT is not to be confused with the Subject "Office Administration" which is an elective.			

In Year 1, a learner must be exposed to a minimum of two skills from the list below, whereas in Year 2 – 4 a learner must be exposed to a minimum of one skill. For continuity, it is recommended that where practicable a learner offers the same skill from Year 2 – 4.

Subjects	Time
Electives	

Agricultural Studies Art and Crafts Civil Technology: Bricklaying and Plastering Civil Technology: Plumbing Civil Technology: Woodworking and Timber Consumer Studies: Food Production Consumer Studies: Needlework Production Early Childhood Development Electrical Technology: Electrical Hospitality Studies Mechanical Technology: Body Works: Panel Beating and or Spray Painting Mechanical Technology: Motor Mechanics Mechanical Technology: Sheet Metal Work Mechanical Technology: Welding Mechanical Technology: Maintenance Office Administration Personal Care: Ancillary Health Care Personal Care: Beauty and Nail Technology Personal Care: Hairdressing Service Technology: Upholstery Wholesale and Retail	13½ hours
Total: General and Occupational	27½

1.5 The table below proposes the learner progression across the years at a School of Skills.

Year 1 Minimum of 1 year of orientation	Year 2	Year 3	Year 4
Base Line Assessment for Language and Mathematics ➤ Intervention (ISP) <u>Fundamental Subjects:</u> • Home Language • FAL • Mathematics • Life Skills: ✓ Personal Social Wellbeing ✓ Physical Education ✓ Creative Arts ➤ <u>ICT Enrichment</u> <u>Electives</u> Minimum 2 x SKILLS Across the year Post Assessment • Analyse results Progress to Year 2 with appropriate support for Languages and Mathematics	<u>Fundamental Subjects:</u> • Home Language • FAL • Mathematics • Life Skills: ✓ Personal Social Wellbeing ✓ Physical Education ✓ Creative Arts ✓ Natural Sciences ➤ <u>ICT Enrichment</u> <u>Electives</u> Minimum of 1 Skill	<u>Fundamental Subjects:</u> • Home Language • FAL • Mathematics • Life Skills: ✓ Personal Social Wellbeing ✓ Physical Education ✓ Creative Arts ✓ Natural Sciences ➤ <u>ICT Enrichment</u> <u>Electives</u> Minimum of 1 Skill	<u>Fundamental Subjects:</u> • Home Language • FAL • Mathematics • Life Skills: ✓ Personal Social Wellbeing ✓ Physical Education ✓ Creative Arts ✓ Natural Sciences ➤ <u>ICT Enrichment</u> <u>Electives</u> Minimum of 1 Skill (External exam- results verified / moderated)

Note:

Year One is an orientation year and learners must be exposed to a minimum of two occupational skills to select a skill with which they will continue from Year Two. Schools that offer more than the minimum two skills in Year One may adapt the Annual Teaching Plan for Year One to accommodate their rotation system to expose learners to more skills e.g. schools may offer a skill per term for Terms 1, 2 and 3 and learners

then select the skill they will specialise in and start it in Term 4. It is important that learners in Year One experience the core competencies of the skills so that an informed choice can be made.

Years Two, Three and Four are the critical years for learners. It is important that learners are exposed to all the Topics and Specific Aims per selected Occupational skill, acknowledging that not all learners will be successful in all of these.

SECTION 2:

INTRODUCTION TO MECHANICAL TECHNOLOGY: WELDING

2.1 What is Welding?

Welding is a fabrication or sculptural process that joins materials, usually metals and other materials, by causing coalescence. This is often done by melting the work pieces and adding a filler material to form a pool of molten material (the weld pool) that cools to become a strong joint, with pressure sometimes used in conjunction with heat, or by itself, to produce the weld.

Many different energy sources can be used for welding, including a gas flame, an electric arc, includes welding (gas, electric, MIG/MAGS), developments, working with sheet metal and the manufacturing of structures. Modern welding techniques includes manual methods like shielded metal arc welding, now one of the most popular welding methods, as well as semi-automatic and automatic processes such as gas metal arc welding.

Metalworking is the process of working with metals to create individual parts, assemblies, or large scale structures. The term covers a wide range of work from large ships and bridges to precise engine parts and delicate jewellery. It therefore includes a correspondingly wide range of skills, processes, and tools. Metalworking is a science, art, hobby, industry and trade. Modern metalworking processes, though diverse and specialized, can be categorized as forming, cutting, or joining processes. Today's machine shop includes a number of machine tools capable of creating a precise, useful work piece.

2.2 Topics to be studied in Welding

1. Safety and first aid – Occupational Health and Safety (OHS) Act;
2. Workshop environment
3. Drawings (rough sketches, simple drawing ISO, Ortho and CAD welding related)
4. Tools and equipment (hand, power and machine)
5. Materials – ferrous and non-ferrous materials
6. Joining methods – permanent and semi-permanent.
7. Welding (operations/procedures and equipment).

2.3 Specific Aims:

The learner is able to:

1. Adhere to and identify safe working practices and demonstrate safe working conditions daily, also adopting proper safety and first aid procedures.

2. Demonstrate knowledge of the welding industry and its productivity requirements, by applying appropriate work-procedures.
3. Understand and interpreting work instructions and drawings for the completion of projects.
4. Apply welding technology, techniques, processes and skills, as applied in the fabrication and welding industry, using appropriate tools and measuring equipment.
5. Identify Ferrous & Non-Ferrous metals the differences, applications and uses of them.
6. Demonstrate the different welding processes: Arc (AC/DC), MIG, TIG, Brazing, Soldering and spot welding.
7. Apply a variety of fillet welding, oxy-fuel cutting and oxy-fuel joining processes.

.2.4 Requirements for Welding as a subject

2.4.1: Time Allocation

The total number of hours allocated for the subject in a five-day cycle is 13 ½ hours. Sufficient time must be allocated in the school timetable for the practical work required to be done.

2.4.2 Resources

Human resources

Welding requires a trained subject specialist (should be a person that has at least 5 years industry experience). It is preferred that the Educator offering Welding is an artisan / technician/ technical educator that has been in the Welding industry.

Welding Educators are required to:

- Teach the subject content with confidence, professionalism and in a relaxed firm manner.
- Plan and execute Theory and Practical lessons/work in a manner that would be suitable for the learners.
- Provide workshop that is properly equipped, clean, safe/first aid friendly for learners.
- Keep proper records of all learners and results from all SBA and PAT.
- Implement innovative methods to keep learners interested in the subject.
- Maintain and service the workshop, tools, and instruments as a whole.
- Produce working PAT projects for learners one for every term and scale at the learner's ability.
- Educator must carry out SBA regularly, at the end of every term correct break down of learner's progress.

Learner Resources:

Each learner should be in possession of the following:

- Workbook.
- Overall.
- Safety steel cap boots.
- Welding helmet.
- Clear safety glasses.
- Leather gloves.
- Tape measure.

2.4.3 Infrastructure, equipment and budget.

Schools must ensure that teachers have the necessary infra-structure, equipment and financial resources for quality teaching and learning.

Infrastructure:

- Welding cannot be implemented in a school without an equipped workshop.
- Electricity supply to the workshop is crucial, preferably a three phase, four-wire supply, but at least single phase with a high current circuit breaker.
- Lighting and ventilation is of extreme importance and a workshop should ideally have multiple exits with doors that open outward.
- Tools and equipment should have sufficient storage and well developed storage management system with an up to date inventory. Shelves should be clearly marked and storage areas defined.
- Good housekeeping principles require that all workshops be cleaned regularly. A suitable waste removal system should be in place to accommodate refuse, off-cut materials as well as chemical waste. The requirements of the Occupational Health and Safety (OHS) Act 85 of 1993 need to be complied with at all times.
- Machinery on stands should be permanently affixed to the floor, with isolation switches for the mains supply. All machines should have working machine guards.
- Electrical motors should ideally be painted bright orange. Specification plates should be clearly legible.
- The workshop must have a lockable mains distribution board. The workshop must be fitted with an emergency cut off switch/s which is/are easily accessible at all times. The red,

mushroom type, emergency switch should preferably be lockable to prevent accidental re-connection with mains in the case of it being activated.

- Safety rules must be displayed on posters in the workshop.

Equipment

SUGGESTED TOOLS AND EQUIPMENT

SAFETY			
MUST HAVE			• Optional
• 1or 2 piece Overall (1/ learner) • Steel capped safety boots (1/ learner) • Safety Glasses (1/ learner)	• Welding helmet (1/ learner) • Leather Apron (1/ learner) • Leather gloves (1/ learner)	• Leather apron (1/ learner) • Ear Muffs (1/ learner)	• Skull cap • Leather anklets • Auto dark welding helmet
• Fire extinguishers 300sqm workshop (5/workshop)	• Sand buckets (3/workshop)	• First Aid kit (wall mounted fully stocked)	
• Full Face clear grinding shield (5/ workshop)		• Water hose on reel (1/workshop)	

TOOLS AND EQUIPMENT			
MUST HAVE			Optional
Hand Tools	Hand Power Tools	Machine Tools	1 of each item per workshop
• Measuring Tools: • Tape Measure (1/ learner) • Steel ruler (5/workshop) • Veneer Callipers (2/ workshop) • Allen keys set (3 sets / workshop)			• Height Gauge • Micro-meter • Laser line gauge
• Marking Out Tools: • Engineering Square Set (1/ workshop) • Scriber (5/workshop) • Boilers Marking Chalk (1box/ Term) • Blue marking ink (2x300ml/ Term) • Sliding bevel (5/workshop) • Callipers: Inside & Outside (1 set) • 200mm Dividers (3/ workshop) • 300mm Dividers (3/ workshop)			• Yellow Paint marker • Chalk line • Spirit level

• Cutting: • Hack saw (1/ learner) • Junior Hack Saw (5/workshop) • Hand held tin snips (5/workshop) • Cole chisel (5/workshop) • Body punches (1 set / workshop) • Hole Saws (1 set / workshop)	• Jig saw (1/ workshop) • 115mm Angle Grinder (15/workshop) • 250mm Angle Grinder (1/ workshop) • Oxy/Acetyl bottles 10kg (1/ workshop)	• Circular band saw (1/ workshop) • Plasma cutter (1/ workshop) • Heavy Duty-Cut Off Saw 450mm (1/ workshop) • Guillotine (1/ workshop) • Portable Band Saw (1/ workshop) • Hydraulic Break press 2.5 meter (1/ workshop)	• Portable Gas Bottles
• Welding: • Positive& Negative welding cables (1set/ machine) • Welding screen 3m (3/ workshop)		• Inverter AC/DC welder (15/workshop) • Core wire MIG welder (MIG welding stands for Metal Inert Gas welding) (2/workshop) • Spot welder (1/ workshop) • TIG welding torch (TIG stands for Tungsten Inert Gas welding) (1/ workshop)	• Oil cooled welder
• Grinding: • 200mm wire brush (1/ learner)	• 150mm angle grinder (15/ workshop)		• Sander on stand • Die grinder

Chipping hammer (1/ learner)	300mm angle grinder (2/ workshop) - Die grinder 250mm (1/ workshop)		
Drilling: - Jobber drill bit set (15/ workshop) - High speed Hole-saw (1 set/ workshop) - Multi stage drill bits (2 set/workshop)	- Hand held drill (2/ workshop) - Rotary SDS held drill (1/ workshop)	Drill press (2 / workshop)	- Cross drill vice - Tap & die set
Joining: 150mm G clamps (3 sets/ workshop) 300mm G clamps (3sets/ workshop) - C clamp grip plier (3sets/ workshop) - Welding magnets (4 sets/ workshop) - Corner jig clamp (2 / workshop) - Rivet gun (2 / workshop)			- Sheet metal clamps - Plate welding plier
Shaping / Forming: - Bench grinder (15/ workshop) - Hand held files (15/ workshop)	Finishing sander (1/ workshop)	100mm Belt sander (1/ workshop)	Orbital sander
Other Accessories:	Radio (1/ workshop)	Extractor fan (1/ workshop)	- Scroll bender - Roll slip bender

• Bench vice (5/ workshop) • Work benches (3/ workshop) • Extension cords 10&20m (2/ workshop) • Screwdrivers set of 8 (3/ workshop) • Hammers-ball pin (1/ learner) • Bolts, nuts and washers • Self- tapers • Two pin adaptor (2/ workshop) • Multi plug adapter (3/ workshop)	• Computer (1/ workshop) • Projector (1/ workshop) • Colour Printer (1/ workshop) • 16 gig Flash drive (1/ workshop) • CAD Software- one of the following • Ally-CAD, Auto-CAD	• Compressor 4 Bar 100lt tank (1/ workshop)	• Sheet metal folder • Compressor • Spray gun & access • Drawing board • 1 Tera-byte Hard Drive
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Consumable items: (minimum requirements)

Hand grit soap	Paper towels	Gas argon, oxygen, acetylene, cougar, LP	Grinding discs 115mm
Welding rods	Drill bits 1-10mm	Grinding discs 115mm and 250 mm	Grinding discs 250mm
Steel brushes	Rags 1/5kg/term	Boiler makers chalk	Masking tape 15,25 & 50mm
Cut-off discs 115-350mm	Flap discs 115mm	Hack saw blades	5kg of waste rags
Sand paper	Wire wheels	Purple scotch brite	Band saw blades
Flap wheels 115mm	Sanding discs 115mm	Sanding discs 250mm	Bench grinder wheel fine grit

Anti-Splatter Spray	Oxy/Act cutting nozzles	Oxy/Act flash Backs	Bench grinder wheel course grit
Clear glass for welding helmet	Green glass for welding helmets	First Aid and safety Box	

Finances:

Budget and inventory

A budget must be allocated for the subject. The amount will be determined by the number of learners taking the subject across all the years and the nature of the practical work required as stipulated in the curriculum. The budget needs to be revised annually and must consider all resources needed per year. The funding must make provision for maintenance of equipment and the replacement over the years.

Resourcing could be sub divided into the following categories:

- ☐ Safety Equipment
- ☐ Tools and Equipment
- ☐ Consumable Materials
- ☐ Practical Assessment Task Resources (PAT)
- ☐ Teaching and Learning Support Material
- ☐ Maintenance

A stock inventory must be maintained by the teacher and verified annually by a Senior Management Team member.

2.5 Career opportunities

Career and occupational opportunities for learners with a foundation in Welding include but is not limited to:

- Continued studies at a college in the NC (V) in a vocational pathway.
- Working as an entrepreneur or working with one.
- Artisan assistant (Semi Skilled);
- Welder (Coded).
- Boiler maker.
- Pressure welder.
- Ship builder.
- Rigger.

- Pattern maker.

SECTION 3:

OVERVIEW OF TOPICS PER TERM AND ANNUAL TEACHING PLANS

3.1 Content overview

TOPIC	Year 1	Year 2	Year 3	Year 4
1. Safety & First Aid	General workshop safety and basic first aid workshop practices in to work area.	Safety and first aid workshop practices HIV/AIDS awareness, machine safety practices	Safety and first aid workshop practices HIV/AIDS awareness, machine safety practice	Safety and first aid workshop practices HIV/AIDS awareness, machine safety practices
2. Communications and Graphics	Implementation of Drawings in the workshop. (rough, ISO and ORTHO)	Implementation of Drawings in the workshop. (rough, ISO and ORTHO)	Communications and Graphics, rough sketches and orthographic drawings	Communications and Graphics, rough sketches and orthographic drawings
3. Workshop environment	Layout of the workshop and the demarcated areas of operation. Cleaning and storage of tools and equipment	Layout of the workshop and the demarcated areas of operation. The proper procedure to cleaning and storage of tools and equipment	Layout of the workshop and the demarcated areas of operation. The proper procedure to cleaning and storage of tools and equipment	Layout of the workshop and the demarcated areas of operation. The proper procedure to cleaning and storage of tools and equipment
4. Tools and equipment	Hand and measuring tools	Hand, measuring tools, portable power Tools	Hand, measuring tools, portable power Tools and all related	Hand, measuring tools, portable power Tools and all related

			equipment for welding	equipment for welding
5. Materials	Origin of steel, profiles	Origin of steel, Profiles and uses	Properties non and ferrous, profiles of metals	Properties and uses metals, profiles of metals
6. Welding operations	All the different types welding equipment, safety and storage. Different types of welding procedures and operations(ARC & GAS)	All the different types welding equipment, safety and storage. Different types of welding procedures and operations(ARC & GAS)	All the different types welding equipment, safety and storage. Different types of welding procedures and operations(ARC GAS, MIG and TIG)	All the different types welding equipment, safety and storage. Different types of welding procedures and operations(ARC, GAS, MIG and TIG)
7. Joining methods	Different types of joining. Demonstrate: A Tack, butt and a fillet weld	Joining method (ARC)	Joining method (ARC and GAS)	Joining method (ARC, GAS MIG and TIG)
8. Entrepreneur			Introduce the Topic of being and Entrepreneur.	Introduce the Topic of being and Entrepreneur.

3.2 Content outlines per term

Year 1

WEE	TOPIC	CONTENT The learner is able to:	Techniques, activities, resources and process notes
1	Introduction: (orientation)	Describe and apply the following: • Workplace layout, • Workplace rules, • Working opportunities in industry	Theoretical and practical explanation to show the Learner the workplace layout while workplace rules are explained. Videos and wall chart information. Worksheet: • Oral discussions in groups • Access to information from reference books and internet resources. • Sort information. • Written and visual presentations. • Practical demonstration. Explanation on the different opportunities that are available in Industry. • Welder. (coded) • Boiler maker. • Grinder. • Vessel maker etc.

2	Safety & First aid	Demonstrate and apply the following safety practises. • Safety procedures, • First Aid boxes; • Protective clothes for head, eye, hands and feet. • Safety signs, colour codes used. Explain and demonstrate how to clean and dress a wound Apply the following conditions: • Unsafe acts & conditions • Good house-keeping • General workshop rules • Demonstrate a workshop is cleaned and explain why. • What unsafe conditions are, the reasons for workshop to be kept safe and clean daily.	Theoretical explanation as well as practical demonstration. Videos and charts to be demonstrated. Worksheets: Oral and Written discussions: • In groups / teams • Reference books and charts with suitable content • Practical activity. Show how to clean and dress a wound properly and to register all injuries in a log book Worksheet and Video content on: • Safe workshop rules and practices. • First aid charts and videos, show what happens in unsafe work conditions. • Introduce the learners to how the workshop is cleaned daily.
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3	Measuring Hand tools	Select use & care: Hand measuring tools. • Tape measure, • Steel ruler, • Meter stick • Veneer Calliper • 90deg square • Sliding adjustable square	Show learners the hand tools how /care and storage of them. Demonstrate the measurement equipment and why it is so important that these tools are taken care of and properly stored away.
4	Hand tools	Identify and describe the caring and demonstrate the correct use and safety of the following hand tools: • Screw Drivers (flat / star) • Hammers • Files (Different shape files) • Spanners • Tin snip • Drill bits • Scriber • Engineers pliers • Hack saw • Vice	Worksheet with picture illustration of hand tools and videos. Use the practical way of describing of hand tools. Hand tools are best taught when busy in a practically demonstrated. Attention should be paid with the safe storage and care of these tools.

5-6	Materials	Identify: Ferrous & Non Ferrous metals (profile & use). Show the physical difference between non and ferrous metals • Brass. • Copper. • Aluminium. • Mild steel. • Stainless steel. • Cast iron. • Copper • Show example and uses of the materials in the workshop.	Show the different profiles that are used that materials are made in. Explain and Demonstrate the different types of materials (mild steel, stainless steel, brass) and the profile. Show example and uses of the materials in the workshop. Learners need to be taken around the school premises and shown where these materials are used. • Stairs cases railings. • Gates at the school. • Support beams. • Window frames.
7-8	Joining	Understand the following • Shielded metal arc welder (components of the machine and use) (AC/DC).	The educator needs to show learners the components of a welder and how the welder operates.

		• Demonstrate all the components of an Arc welder and the proper way to connect it before use. • The different colours that is used to for the identification of positive and earth cables. • Do a tack weld on measured material	Use the internet/media to show visually what welding processes are available. Demonstrate to learners difference between AC/DC welders: • Define AC (Alternating Current). • Define DC (Direct Current). Educator should demonstrate to learners how to do a tack weld correctly on two pieces of properly prepared material. But this must be mechanically strong not to break easily.
9-10	Assessment	The weeks allocated for formal assessment are integrated across the weeks planned for teaching and learning. The assessment will consist of Practical Task/s with a 75% weighting and a Theory test with a 25% weighting.	

Activity 1- Practical Demonstration (Measuring, cutting and Tack Welding) = 25% of the Term mark.

The Learners must be able to demonstrate certain skills acquired during the term. This is in the form of a single task and not part of the Terms practical model.

Learner measures, cuts and prepare two pieces of material as per the drawing. The learner must produce two tack welds that should be mechanically strong. All three stages must be assessed.

- Measuring
- Mark off
- Cutting

- Cleaning off sharp edges
- Tack Weld

Total for Activity 1 Marks to be converted to 25% of the total term mark.

Activity 2 – PRACTICAL ASSESSMENT TASK – PAT (Practical Project) = 50% of term mark.

Fundamental knowledge = 25%

Practical learning areas = 25%

Learners must now produce a practical project (PAT) using the all the method of construction (in Weeks 1 – 10) which can be assessed to evaluate their acquisition skills. This project should involve the skills, techniques and knowledge of the theory component.

Assessments are done according to a rubric which includes all the skills acquired during the term.

- Examples of projects are in the Addendum.

Activity 3 – Formal written or Oral Theory assessment = 25%.

Scope of work that is done during the term.

Annexure A: Theory Assessment: Oral and /or Written Pages

Theory section: This section comprises of all the above work done in weeks 1 through 10 inclusions should be pictures and questions with a multi answer block. Pictures should constitute a major part of this section.

Annexure B: Practical Assessment: Page

Practical section: learners are now assessed on their understanding of safety, standards and practices as well as measurement which forms the basis of the all practical work done in the workshop. Learners in the first year should be able complete a model (which should be at their level, but the ability to finish the model.

WEEK	TOPIC	CONTENT The learner is able to:	Techniques, activities, resources and process notes
1	Introduction: (orientation)	Understand the Welding concept and the work place layout. Enter and exit the workshop. As well as observe the safety procedure with entering and exiting a workshop. - Identify the safety signs that indicate the doors used for evacuation. - Explain the safety evacuation drill and perform it.	Explain the Welding definition to the learners and orientate them on the entrance and exits of the workshop. - Videos about safety. - All exits and entrances should be marked and explained. - Demonstrate in a class group how to evacuate the workshop using a proper evacuation drill. - Written presentations - Sort information.
2	Safety, first aid HIV and AIDs (OHS Act)	Demonstrate knowledge for the following - Safety procedures, - First Aid boxes. - Protective clothes, - Safety signs.	Theoretical explanation as well as practical demonstration. Videos and charts to be demonstrated. Worksheets: Oral and Written discussions: In groups / teams

		Learners need to explain the concept and application of how to clean and dress a wound. Explain the following conditions: • Personal safety (PPE – Personal Protective Equipment) • Unsafe acts & conditions • Good house-keeping • General workshop rules • Demonstrate how a workshop is cleaned and explain why. • What unsafe conditions are, the reasons for workshop to be kept safe and clean daily. Explain the proper protocol for dealing with victims with HIV and AIDs. • Wearing gloves • Protective eye wear • Checking yourself for no open wounds • Treating the HIV and AIDs victim with caution • Proper disposal of all cleaning materials	• Reference books and charts with suitable content • Practical activity. Role play emergency procedures. Show how to clean and dress a wound properly and to register all injuries in a log book Worksheet and Video content on : • Safe workshop rules and practices. • First aid charts and videos, show what happens in unsafe work conditions. • Introduce the learners to how the workshop is cleaned daily. Educator to demonstrate the HIV and Aids protocol: • Videos and charts • Oral discussions • Worksheets • Present information visually • Seek information from nursing sister.
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3-4	Equipment: Measuring Hand tools	- Identify measuring tools and the proper use thereof. - Accurately measure length, width and thickness in mm Select use & care: Hand measuring tools. - Tape measure, - Steel ruler. - Meter stick. - Veneer Calliper. 90deg square. - Sliding adjustable square. - Scriber and Marking gauge. Explain and demonstrate the proper procedure when drilling a hole into material. The five point step should be put into practice	Educator with the aid of theory show learners hand tools and how to care and storage of them. Demonstrate the measurement equipment and why it is so important that these tools are taken care of and properly stored away. - Demonstrate how to measure and read in mm. - Educate how to use the measuring equipment. - Demonstrate how important accuracy is. - Explain how proper right angles are obtained. - Show learners modern technology advancements. The educator must explain and demonstrate the 5 point procedure of how a hole should be drilled into material. - Measure and mark the spot. - Centre-punch the spot. - Drill a pilot hole. - Drill the hole that is required (finish hole). - Chamfer the holes both sides to remove all sharp edges.
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5-6	Materials	Identify: Ferrous metals and non ferrous metals (profile & use). Show the physical difference between non- ferrous and ferrous metals. Identify the materials that they use most frequently. • Brass, • Copper • Aluminium • Mild steel • Stainless steel • Cast iron. • Copper • Show example and uses of the materials in the workshop.	Theoretical explanation of the topic: Show the different profiles that are used that materials are made in. Educator can explain and use a magnet to show the difference between non-ferrous and ferrous materials. Explain and Demonstrate the different types of materials. • Theory can be explained while giving a practical demonstration. • Spend more time explaining the material differences as this will give the learner the proper information when choosing the appropriate material. • Provide the learner with the notes and descriptions of these materials. Show example and uses of the materials in the workshop.

7-8	Joining methods (tack and fillet weld)	Explain the following types of joining methods: • Riveting • Use nuts and bolts. • Welding. • Prepare material before a tack weld. • Perform a tack weld. • Perform a fillet weld no longer than 30mm. • Clean and present a fillet weld.	Use practical demonstrations to show learners the components that should be used to complete a successful joint: Educators explanation and demonstration for Riveting: • Prepare material before riveting. • The correct holes to be drilled for riveting. • The riveting procedure. Educators explanation and demonstration for use of nuts and bolts method: • Prepare material before nuts and bolts to be used. • The correct holes to be drilled for bolts. • The nuts and bolts tightening procedure. • The correct nuts and bolts to be used (mechanical strength). Educators explanation and demonstration for Welding method:

			• Prepare material before welding. • Tack weld • Fillet weld • Clean weld off to present. Both must have mechanical strength to be classified as the joints listed.
9 – 10	Formal Assessment	The weeks allocated for formal assessment are integrated across the weeks planned for teaching and learning. The assessment will consist of Practical Task/s with a 75% weighting and a Theory test with a 25% weighting. Activity 1- Practical Demonstration (Measuring, cutting and Tack Welding) = 25% of the Term mark. The Learners must be able to demonstrate certain skills acquired during the term. This is in the form of a single task and not part of the Terms practical model. Learner measures, cuts and prepares two pieces of material as per the drawing. The learner must produce two tack welds that should be mechanically strong. All three stages must be assessed. • Measuring • Cutting • Cleaning off sharp edges • Tack Weld • Fillet Weld Total for Activity 1 Marks to be converted to 25% of the total term mark. Activity 2 – PRACTICAL ASSESSMENT TASK – PAT (Practical Project) = 50% of term mark.	

Fundamental knowledge = 25%

Practical learning areas = 25%

Learners must now produce a practical project (PAT) using the all the method of construction (in Weeks 1 – 10) which can be assessed to evaluate their acquisition skills. This project should involve the skills, techniques and knowledge of the theory component.

Assessments is done according to a rubric which includes all the skills acquired during the term. Marks total 50% of the term.

- Examples of projects are in the Addendum.

Activity 3 – Formal written or Oral Theory assessment = 25%.

Scope of work that is done during the term.

Assessment: consists of two sections theory 25% and practical 75% = 100%

Annexure C: Theory Assessment: Oral and /or Written Pages

Theory section: (oral or written) score allocation 25%: comprises of all the above work done in weeks 1 through 10 inclusions should be pictures and multi answer questions. Pictures should constitute a major part of this section.

Annexure D: Practical Assessment: Pages

Practical section: Learners should now be ready to for an assessment with a use of hand tools so that the theoretical aspect can be implemented. Measurement now plays a role as this is the foundation to which all work can commence in the workshop.

At the end of this term: A project should be completed as evidence of the learner's ability.

(NB! Not all learners will be able to produce an exact reproduction of the example the educator has provided)

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WEEK	TOPIC	CONTENT The learner is able to:	Techniques, activities, resources and process notes
1-2	Re-Cap Safety and Apply work site practices	Demonstrate knowledge for the following • Safety procedures, • First Aid boxes. • Protective clothes, • Safety signs. Explain the application of how to clean and dress a wound. Explain the following conditions: • Personal safety(PPE – Personal Protective Equipment) • Unsafe acts & conditions • Good house-keeping • General workshop rules • Demonstrate a workshop is cleaned and explain why.	Theoretical explanation as well as practical demonstration. Videos and charts to be demonstrated. Worksheets: Oral and Written discussions: • In groups / teams • Reference books and charts with suitable content • Practical activity. • Role plays emergency procedures. Show how to clean and dress a wound properly and to register all injuries in a log book Worksheet and Video content on: • Safe workshop rules and practices. • First aid charts and videos, show what happens in unsafe work conditions. • Introduce the learners to how the workshop is cleaned daily.

		• What unsafe conditions are, the reasons for workshop to be kept safe and clean daily. Explain the application and the proper protocol for dealing with victims with HIV and AIDs. • Wearing gloves • Protective eye wear • Checking yourself for no open wounds • Treating the HIV and AIDs victim with caution • Proper disposal of all cleaning materials	Educator to demonstrate the HIV and Aids protocol: • Videos and charts • Oral discussions • Worksheets • Present information visually • Seek information from nursing sister.
3- 5	Equipment: Hand tools Power Hand Tools	Explain the selection, use and care for engineering hand tools and hand power tools. Hand tools examples: • G clamps. • Welding clamps. • Long nose pliers. • Hack saw and junior. • Wire cutters. • Side cutters.	Theoretical explanation and demonstrations of the topics: • The theory can be explained while doing a practical demonstration but pen on paper is important. • Provide the learner with the appropriate note where textbooks are not available. • Attention must be given to correct terminology, safe use and proper care of tools is required.

		• Vice grips. • Bolt cutters. • Spanners and sockets. • Hand Files. Power hand tools examples: • 115mm angle grinder. • Hand drill 13mm chuck. • Jig saw. • Die grinder. • 250mm grinder. • SDS rotary drill. • 300mm cut of saw. • Bench grinder 200mm. Explain the following: • Repair and replace a faulty 3 core cable. • Repair and replace a faulty 3 pin plug. • Store and clean hand power tools. • Proper care and storage of extension leads.	• All topics and content should be covered in theory. Demonstrate the correct way in which power tools need to be stored and cared for. The correct way by demonstrating the care repair and storage of electrical cabling. • How to replace a faulty cable on a hand power tool, following the correct procedures • How to replace a 3pin plug. • How to replace a switch and carbon brushes.
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6,7	Materials	Demonstrate sound understanding of • Ferrous and Non Ferrous materials • The difference applications where they are used. • Shelf space for materials • Care a preventative maintenance for materials to be stored. • Explain how mild steel and galvanised materials need to be store separately.	Theoretical identification of materials in accordance with material specification sheet as included in work instructions. Educator should demonstrate the following tests for Ferrous and Non ferrous metals. Wall charts and internet information can be used to show the learners these applications. • The magnetic test. • The spark test Display different materials used in industry. Demonstrate all safety precautions in relation to work site practices and procedures. On the storage of materials an why they need to be stored separately.
8	Welding symbols and joints	Explain the following: • Know the basics of welded joints and their terminology • Sketch and describe welding symbols used in the industry	Explain and describe weld symbols used in the industry in accordance with AWS instruction manual. Welding symbols is limited to those as prescribed in the AWS Standards and practices. • Rough sketch of the symbol for a fillet joint

		- Describe and sketch weld symbols in accordance with AWS and work instruction sheet. - Sketches of weld symbols can be done free hand or formal drawing format.	Symbol for a lap joint
9 – 10	Formal Assessment	The weeks allocated for formal assessment are integrated across the weeks planned for teaching and learning. The assessment will consist of Practical Task/s with a 75% weighting and a Theory test with a 25% weighting. Activity 1- Practical Demonstration (Measuring, cutting, welding, grinding and drilling) = 25% of the Term mark. The Learners must be able to demonstrate certain skills acquired during the term. This is in the form of a single task and not part of the Terms practical model. A learner measure cut and prepares two pieces of material as per the drawing. The learner must produce tack welds that should be mechanically strong. A fillet weld should be performed at least 30mm, a hole needs to be drilled and accuracy is really important. All five stages must be assessed. - Measuring - Cutting - Welding - Drilling - Grinding Total for Activity 1 Marks to be converted to 25% of the total term mark. Activity 2 – PRACTICAL ASSESSMENT TASK – PAT (Practical Project) = 50% of term mark.	

Fundamental knowledge = 25%

Practical learning areas = 25%

Learners must now produce a practical project (PAT) using the all the method of construction (in Weeks 1 – 10) which can be assessed to evaluate their acquisition skills. This project should involve the skills, techniques and knowledge of the theory component.

An assessment is done according to a rubric which includes all the skills acquired during the term. Marks total 50% of the term.

- Examples of projects are in the Addendum.

Activity 3 – Formal written or Oral Theory assessment = 25%.

Scope of work that is done during the term.

Assessment: consists of two sections theory 25% and practical 75% = 100%

Annexure C: Theory Assessment: Oral and /or Written Pages

Theory section: (oral or written) score allocation 25%: comprises of all the above work done in weeks 1 through 10 inclusions should be pictures and multi answer questions. Pictures should constitute a major part of this section.

Annexure D: Practical Assessment: Pages 61

Practical section: Learners should now be ready to for an assessment with a use of hand tools so that the theoretical aspect can be implemented. Measurement now plays a role as this is the foundation to which all work can commence in the workshop.

At the end of this term: A project should be completed as evidence of the learner's ability.

(NB! Not all learners will be able to produce an exact reproduction of the example the educator has provided)

WEEK	TOPIC	CONTENT The learner is able to:	Techniques, activities, resources and process notes
1-2	Re-cap Safety and procedures. Re-Cap:- Welding symbols and joints	Explain all safety and First Aid, HIV and AIDS procedures. Know the basics of welded joints and their terminology Sketch and describe welding symbols used in the industry. Describe and sketch weld symbols in accordance with AWS and work instruction sheet. Sketches of weld symbols can be done free hand or formal drawing format.	Explain all safety, first aid HIV and Aids procedures, learners need to be shown all the signs and pictures to establish that safety should be first priority in the workshop. Use the projector and signs in the workshop. Recap: Welding symbols is limited to those as prescribed in the AW Standards and practices.
3-4	Equipment: Hand tools and Power Hand Tools	Explain and understand how to select, use and care for engineering hand tools and hand power tools. Hand tools examples: • Circle divider • Bench vice	Theoretical explanation and demonstrations of the topics: • The theory can be explained while doing a practical demonstration but pen on paper is important. • Provide the learner with the appropriate note where textbooks are not available.

		• Rubber mallet • Vice grips • Cole chisel • Allen and torque keys • Wire strippers • Circlip pliers inner and outer • Tin snips Power hand tools examples: • Nibbler • Portable hand drill • Sander • Heat gun • Vacuum cleaner Do the following: • Repair and replace a faulty 3 core cable. • Repair and replace a faulty 3 pin plug. • Store and clean hand power tools.	• Attention must be given to correct terminology, safe use and proper care of tools is required. • All topics and content should be covered in theory. Demonstrate the correct way in which power tools need to be stored and also the way electrical cabling needs to be properly rolled up.
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5	Drawings	Explain and understand the following: • Measure, • estimate and • calculate physical quantities and • explore, describe and • represent geometrical relationships in 2-dimensions in different life or workplace context Explore transformations of two-dimensional geometric figures • Square • Cone • Rectangle • Cube • Sphere • Triangle • Circle	Theoretical explanation by the Educator and to demonstrate and correctly read scales on the measuring instruments. • Identify the appropriate instrument to measure a particular quantity. Carry out calculations correctly. • Explain and demonstrate appropriate units measurement and calculation. • Describe and identify properties of symmetrical shapes. • Explain and show concepts of lines of symmetry in 2-dimensional figures using paper folding and reflections in the lines of symmetry. • Demonstrate how learners are able to cut-out paper figures and assemble them into the required shapes that are needed. Show learners these shapes by means of chart diagrams and video link.
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6-8	Welding process Arc Welder Equipment	Describe the shielded metal arc welding process Explain and demonstrate the importance of correct assembly of the shielded metal arc welding (SMAW) equipment, and the consequences of incorrect assembly, with reference to the vendor requirements. Select, set up and conduct pre-operational checks of shielded metal arc welding equipment: Identify and connect an Arc weld for operation: • Suitable power source • Earth clamp • Electrode holder • Welding settings. Proper care of:	Educator should theoretically explain terms and definitions used with welding terminology as recorded in international welding standards. How to repair broken cables and to maintain the equipment. • Demonstrate the correct way to connect welding machine. • The correct polarity connections. • How to maintain equipment. • The right equipment to use before you start to weld. Carry out pre-operational checks in accordance with manufacturer's specifications. Resources to include: manufacturer's operational manual, worksite practices and safety and Demonstrate verify and selection shielded metal arc welding equipment as specified on welding procedure specification environmental issues. The correct amperage to be used to weld different gauge materials. The correct size welding rods to be used for different types of material.
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		• Welding helmet • Safety glasses • Leather gloves. • Leather apron	Identify and rectify hazards relating to welding process in accordance with standard work site practices.
9 – 10	Formal Assessment	The weeks allocated for formal assessment are integrated across the weeks planned for teaching and learning. The assessment will consist of Practical Task/s with a 75% weighting and a Theory test with a 25% weighting. Activity 1- Practical Demonstration (Measuring, cutting, welding, grinding and drilling) = 25% of the Term mark. The Learners must be able to demonstrate certain skills acquired during the term. This is in the form of a single task and not part of the Terms practical model. Learner measures, cuts and prepares two pieces of material as per the drawing. The learner must produce tack welds that should be mechanically strong. A fillet weld should be performed at least 30mm, a hole needs to be drilled and accuracy is really important. All five stages must be assessed. • Measuring • Cutting • Welding • Drilling • Grinding Total for Activity 1 Marks to be converted to 25% of the total term mark. Activity 2 – PRACTICAL ASSESSMENT TASK – PAT (Practical Project) = 50% of term mark. Fundamental knowledge = 25%	

Practical learning areas = 25%

Learners must now produce a practical project (PAT) using the all the method of construction (in Weeks 1 – 10) which can be assessed to evaluate their acquisition skills. This project should involve the skills, techniques and knowledge of the theory component.

Assessments are done according to a rubric which includes all the skills acquired during the term. Marks total 50% of the term.

- Examples of projects are in the Addendum.

Activity 3 – Formal written or Oral Theory assessment = 25%.

Scope of work that is done during the term.

Assessment: consists of two sections theory 25% and practical 75% = 100%

Annexure C: Theory Assessment: Oral and /or Written Pages

Theory section: (oral or written) score allocation 25%: comprises of all the above work done in weeks 1 through 10 inclusions should be pictures and multi answer questions. Pictures should constitute a major part of this section.

Annexure D: Practical Assessment: Pages 61

Practical section: Learners should now be ready to for an assessment with a use of hand tools so that the theoretical aspect can be implemented. Measurement now plays a role as this is the foundation to which all work can commence in the workshop.

At the end of this term; A project should be completed as evidence of the learner's ability.

(NB! Not all learners will be able to produce an exact reproduction of the example the educator has provided).

WEEK	TOPIC	CONTENT The learner is able to:	Techniques, activities, resources and process notes
1 - 2	Re-cap Safety and procedures and Welding operations	Explain all safety and First Aid, HIV and AIDS procedures. Explain the following: • Select, set up and conduct pre-operational checks of shielded metal arc welding equipment. • Identify, verify and selection shielded metal arc welding equipment as specified on welding procedure specification. • Prepare work pieces prior to welding as specified on drawing and worksite procedures.	Explain all safety, first aid HIV and Aids procedures, learners need to be shown all the signs and pictures to establish that safety should be first priority in the workshop. Use the projector and signs in the workshop. Educator must practically demonstrate and rectify hazards relating to welding process Carry out pre-operational checks in accordance with manufacturer's specifications. • Resources to include: manufacturer's operational manual, worksite practices and safety and environmental issues in accordance with standard work site practices. Tack weld work-pieces in position as per drawing specifications. • The right material for the operation. • The correct procedures and tools for the manufacture of the application. • G clamps, C clamps, jigs.

3 - 4	Welding gas arc process Core welding process	Describe the gas metal arc welding process and related equipment Explain the importance of correct assembly of the gas metal arc welding equipment, and the consequences of incorrect assembly, with reference to the manufacturer's requirements. Describe and demonstrate the cored-wire welding process.	Educator must explain and demonstrate welding parameters in accordance with job requirements. <u>Resources to include:</u> manufacturer's specification, worksite practices and safety and environmental issues. Welding parameters: volts/amps, gas-flow, wire speed, and wire stick out as specified in the welding procedures specifications. Explain terms and use definitions consistent with generally accepted welding terminology as recorded in international welding standards; Suitable power source, wire-feeder, shielding gas, regulator, flow-meter, materials as specified on drawings and weld filler material, according to welding procedure specifications. Explain and demonstrate work-pieces prepared prior to welding as specified on drawing and worksite procedures, together with safety aspects. • Show learners as per internet pictures what a tack weld looks like.
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			Inspect work-piece prior to welding. Check work piece that it is properly tack welded and that the tack weld is strong enough to hold the pieces together. Demonstrate safety precautions. Prepare work-pieces prior to welding. • Demonstrate tack weld work-piece in position as per drawing specifications. Check dimensions and alignment as specified on drawing Resources include: • Worksite procedures, tools, equipment, safety requirements, and materials.
5	Calculations	• Use mathematics to plan and control projected budgets and income and expenditure. • Use a calculator • Book and pen • Use a drawing as reference for costing.	Theoretical explanations of the following should be done by the educator. All calculations using a calculator efficiently and correctly, and solutions obtained are verified in terms of the context. Explain projected income and expenditure realistically. Demonstrate: • Material to be used and cost

			• Drawing to be used • Funds available for project • Cost of making a project • Show the profit made by completing the project.
6-7	Recap all the topics work done for the year	Revision the following concepts • Safety, • First aid, • Tools, • Materials • Joining. • Recap and reflecting: this will make sure that learners are properly prepared for joining methods. • Drawings. • Welding processes.	The educator must revise and reflecting, together with all relevant and related materials: • Wall charts, • Notes • Theoretical notes • Power point presentations Recap and reflecting: together with all relevant and related materials, wall charts, notes and power point presentations. Recap and reflecting: together with all relevant and related materials, wall charts, notes and power point presentations • Welding operations. • Gas arc processes. • Calculations.
8 – 10	Formal Assessment	The weeks allocated for formal assessment are integrated across the weeks planned for teaching and learning. The assessment will consist of Practical Task/s with a 75% weighting and a Theory test with a 25% weighting.	

Activity 1- Practical Demonstration (Measuring, cutting, welding, grinding and drilling) = 25% of the Term mark.

The Learners must be able to demonstrate certain skills acquired during the term. This is in the form of a single task and not part of the Terms practical model.

Learner measures, cuts and prepares two pieces of material as per the drawing. The learner must produce tack welds that should be mechanically strong. A fillet weld should be performed at least 30mm, a hole needs to be drilled and accuracy is really important. All five stages must be assessed.

- Measuring 20% (5)
- Cutting 20% (5)
- Welding 20% (5)
- Drilling 20% (5)
- Grinding 20% (5)

Total for Activity 1 Marks to be converted to 25% of the total term mark.

Activity 2 – PRACTICAL ASSESSMENT TASK – PAT (Practical Project) = 50% of term mark.

Fundamental knowledge = 25%

Practical learning areas = 25%

Learners must now produce a practical project (PAT) using the all the method of construction (in Weeks 1 – 10) which can be assessed to evaluate their acquisition skills. This project should involve the skills, techniques and knowledge of the theory component.

Assessments are done according to a rubric which includes all the skills acquired during the term. Marks total 50% of the term.

- Examples of projects are in the Addendum.

Activity 3 – Formal written or Oral Theory assessment = 25%.

Scope of work that is done during the term.

WEEK	TOPIC	CONTENT The learner is able to:	Techniques, activities, resources and process notes
1	Revision of Safety, first aid HIV and AIDs (OHS Act	Demonstrate understanding of the following • Safety procedures, • First Aid boxes; • Protective clothes • Colours that signs are made out off and meaning. • Safety signs. Clean and dress a wound Apply the following conditions: • Personal safety (PPE – Personal Protective Equipment) • Unsafe acts & conditions • Good house-keeping • General workshop rules • Demonstrate a workshop is cleaned and explain why.	Theoretical explanation as well as practical demonstration. Videos and charts to be demonstrated. Worksheets: Oral and Written discussions: • In groups / teams • Reference books and charts with suitable content • Practical activity. Role play emergency procedures. Show how to clean and dress a wound properly and to register all injuries in a log book Worksheet and Video content on: • Safe workshop rules and practices. • First aid charts and videos, show what happens in unsafe work conditions. • Introduce the learners to how the workshop is cleaned daily.

		• What unsafe conditions are, the reasons for workshop to be kept safe and clean daily. Apply the proper protocol for dealing with victims with HIV and AIDs. • Wearing gloves • Protective eye wear • Checking yourself for no open wounds • Treating the HIV and AIDs victim with caution • Proper disposal of all cleaning materials.	Educator to describe the HIV and Aids protocol: • Videos and charts • Oral discussions • Worksheets • Present information visually • Seek information from nursing sister.
2	Communications & Graphics	A learner will demonstrate the following: • Explain importance of graphics / drawing in the workshop. • Able to do rough / freehand drawings. • Scale up drawing to 1:1. • Do a material costing of a workshop project. Understand the importance of the proper way to cost a Job / project.	Educator: Introduction of Communications and Graphics. • Practical role of graphics / drawings. • Explain the difference scale to drawings. • Demonstrate rough drawings. • Orthographic drawings. • Introduce learners to costing and profit making.

3-4	Equipment: Measuring Hand tools	Explain the following: • Identify measuring tools and the proper use thereof. • Accurately measure length, width and thickness in mm. Select use & care: Hand measuring tools. • Tape measure, • Steel ruler, • Meter stick • Venier Calliper • 90deg square • Sliding adjustable square • Height gauge with marble top.	Educator with the aid of theory show learners hand tools and how to care and storage of them. Demonstrate the measurement equipment and why it is so important that these tools are taken care of and properly stored away. • Demonstrate how to measure and read in mm. • Educate how to use the measuring equipment. • Demonstrate how important accuracy is. • Explain how proper right angles are obtained. • Show learners modern technology advancements.
5-6	Materials	Explain the following: • Identify: Ferrous & non- ferrous metals (profiles & use). • Show the physical difference between the two and the different applications where and why they are used. • Brass,	Practical demonstration by the educator for the following and the proper tests to be done to show the differentiation between non and ferrous metals.

		• Copper • Aluminium. • Mild steel. • Stainless Steel. • Cast iron.	The educator must do the following tests and should be demonstrated with the aid of wall charts and internet. • The magnetic test. (magnet to be used) • The spark test. (grinding wheel for specific sparks) • Show the different profiles that are used that materials are made in. • Explain and demonstrate the different types of materials • Profiles of the materials. • The different uses they have in the daily use in industry and in homes and why they needed to be used in all situations.
7-8	Welding process Arc Welder Equipment	Describe the shielded metal arc welding process (SMAW). Explain and demonstrate the importance of correct assembly of the shielded metal arc welding equipment, and the consequences of incorrect assembly, with reference to the vendor requirements.	Educator should theoretically explain terms and definitions used with welding terminology as recorded in international welding standards. How to repair broken cables and to maintain the equipment. Demonstrate the correct way to connect welding machine.

		Select, set up and conduct pre-operational checks of shielded metal arc welding equipment: Explain identify and connect an Arc weld for operation: • Suitable power source single and 3 phase • Earth clamp • Electrode holder • Welding settings. • Welding rod specifications for the appropriate job. • Maintenance of cables and helmets. • Proper storage of welding equipment and cables. Proper care of: • Welding helmet • Safety glasses • Leather gloves. • Leather apron.	• The correct polarity connections. • How to insert the welding rod properly in its holder. • The right equipment to use before you start to weld Carry out pre-operational checks in accordance with manufacturer's specifications. Resources to include: manufacturer's operational manual, worksite practices and safety. Demonstrate verify and selection shielded metal arc welding equipment as specified on welding procedure specification environmental issues. The correct amperage to be used to weld different gauge materials. The correct size welding rods to be used for different types of material. Identify and rectify hazards relating to welding process in accordance with standard work site practices.
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9 – 10	Formal Assessment	The weeks allocated for formal assessment are integrated across the weeks planned for teaching and learning. The assessment will consist of Practical Task/s with a 75% weighting and a Theory test with a 25% weighting. Activity 1- Practical Demonstration (Measuring, cutting, welding, grinding and drilling) = 25% of the Term mark. The Learners must be able to demonstrate certain skills acquired during the term. This is in the form of a single task and not part of the Terms practical model. A certain degree of difficulty needs to increase from term to term and the learner's ability should be taken into consideration. Learner measures, cuts and prepares two pieces of material as per the drawing. The learner must produce tack welds that should be mechanically strong. A fillet weld should be performed at least 30mm, a hole needs to be drilled and accuracy is really important. All five stages must be assessed. • Measuring • Cutting • Welding • Drilling • Grinding Total for Activity 1 Marks to be converted to 25% of the total term mark. Activity 2 – PRACTICAL ASSESSMENT TASK – PAT (Practical Project) = 50% of term mark. Fundamental knowledge = 25% Practical learning areas = 25% Learners must now produce a practical project (PAT) using the all the method of construction (in Weeks 1 – 10) which can be assessed to evaluate their acquisition skills. This project should involve the skills, techniques and knowledge of the theory component.
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Assessments are done according to a rubric which includes all the skills acquired during the term. Marks total 50% of the term.

- Examples of projects are in the Addendum.

Activity 3 – Formal written or Oral Theory assessment = 25%. Scope of work that is done during the term.

Practical:

Activity 1: Fundamental section 25%. The learners should firstly select all safety equipment before starting assessment. Select and identify all the tools need to complete the project. Understand from the assessment worksheet what needs to be done to complete the assessment.

Activity 2: Practical Section 50%. Measure and mark the material to be used. Follow the Assessment worksheet to complete the project to be made.

Theory:

Activity 3: Theory section comprises of 25%. This is an oral/ written section whichever the Learner is comfortable with. The answer block section helps the learners to choose the correct answer as well as the correct identification of colour pictures of related equipment.

WEEK	TOPIC	CONTENT The learner is able to:	Techniques, activities, resources and process notes
1	Revision of Safety, first aid HIV and AIDs (OHS Act	Demonstrate knowledge for the following • Safety procedures, • First Aid boxes; • Protective clothes • Colours that signs are made out off and meaning. • Safety signs. Clean and dress a wound Apply the following conditions: • Personal safety (PPE – Personal Protective Equipment) • Unsafe acts & conditions. • Good house-keeping. • General workshop rules. • Demonstrate a workshop is cleaned and explain why. • What unsafe conditions are, the reasons for workshop to be kept safe and clean daily.	Theoretical explanation as well as practical demonstration. Videos and charts to be demonstrated. Worksheets: Oral and Written discussions: • In groups / teams • Reference books and charts with suitable content • Practical activity. • Role plays emergency procedures. Show how to clean and dress a wound properly and to register all injuries in a log book Worksheet and Video content on: • Safe workshop rules and practices. • First aid charts and videos show what happens in unsafe work conditions.

		Apply the proper protocol for dealing with victims with HIV and AIDs. • Wearing gloves. • Protective eye wear. • Checking yourself for no open wounds. • Treating the HIV and AIDs victim with caution. • Proper disposal of all cleaning materials.	• Introduce the learners to how the workshop is cleaned daily. Educator to demonstrate the HIV and Aids protocol: • Videos and charts • Oral discussions • Worksheets • Present information visually • Seek information from nursing sister.
2	Graphics & Communications	Measure, estimate and calculate physical quantities and explore, describe and represent geometrical relationships in 2-dimensions in different life or workplace context Estimate, measure and calculate physical quantities to solve problems in practical situations. Estimate, measure and calculate physical quantities to solve problems in practical situations.	Correctly read scales on the measuring instruments and identify the appropriate instrument to measure a particular quantity. Carry out calculations correctly. Explain and demonstrate appropriate units measurement and calculation. Describe and identify properties of symmetrical shapes. Explain and identify concepts of lines of symmetry in 2-dimensional figures using paper folding and reflections in the lines of symmetry.

			Explain the concepts of transformation in terms of reflections, translations and rotations using concrete materials.
3-4	Equipment:	Explain the selection, use and care for engineering hand tools and hand power tools. Hand tools examples: • Dividers • G clamps • Cole chisel • Various files and uses • Scroll benders • Ball peen hammer • Anvil • Spanners and sockets. • Jigs. • Hand scrollers. • Metal twistors. • Metal shearing. Power hand tools examples: • Compressor • Pneumatic hand tools • Hand drill	Educator to theoretical explanation and demonstrations of the topics: • The theory can be explained while doing a practical demonstration but pen on paper is important. • Provide the learner with the appropriate note where textbooks are not available. • Attention must be given to correct terminology, safe use and proper care of tools is required. • All topics and content should be covered in theory. Educator to demonstrate the correct way in which power tools need to be stored and also the way electrical cabling needs to be properly rolled up.

		• Finishing grinder • Heat gun • SDS rotary drill. Floor mounted equipment: • Folder • Bender • Bench grinder • Guillotine. • Pedestal drill. • 300mm diameter disc sander. • Repair and replace a faulty 3 core cable. • Repair and replace a faulty 3 pin plug. • Store and clean hand power tools.	Educator to do theoretical explanation and demonstrations of the following: • Repairs to power cables. • Replacing power cables • Replacing the plug top • Replacing brushes and bearings inside power hand tools and motors.
5-6	Material selection	Explain, Identify and select material to specification. Plan the identification process to identify materials to specification on. Explain and demonstrate the steps in the identification of materials to specification to be in logical sequence and to job requirements.	Educator to demonstrate all safety precautions in relation to work site practices and procedures. Educator needs to explain and demonstrate the following tests with the aid of wall charts and internet. • The magnetic test. (use a magnet)

		Identification of materials in accordance with material specification sheet as included in work instructions. Explain the following joints: • Butt joint. • T joint • Lap joint	• The spark test. (use a grinding wheel for spark specifics) Identify materials marked with colour code or identification numbers. • Materials to be identified is limited to mild steel, aluminium, stainless steel and carbon steels. Identification methods visually with the aid of de-coding systems and mechanical processes. Educator demonstrates and explains with wall charts, video and internet the following joints that are widely used in industry: • Butt joint. • T joint. • Lap joint.
8	Entrepreneurship (Career Opportunities)	Learners understand the processes involved in becoming an Entrepreneur Requirements of becoming an Entrepreneur.	Introduce and explain the processes of becoming an Entrepreneur. There are Critical Stages You will go through as an Entrepreneur:

		Essential Skills Required to Succeed as An Entrepreneur • Ambition. It is easy to give up when the going gets tough, but the most successful entrepreneurs persist because of their ambitious nature. ... • Willingness to Learn. ... • Ability to Listen. ... • Creativity. ... • Assertiveness and Confidence. ... • Perseverance. ... • Courage and Risk Taking.	• Finding The Right Business Idea: ... • Developing the business plan: ... • Raising Your Seed Funding: ... • Getting Paying Customers: ... • Success or Maybe Failure: Documentation for becoming an Entrepreneur (South African Citizen) • South African Citizen Identity Number. • South African Citizen Income Tax Number. • Proof of Residential Address. • Comprehensive Curriculum Virtea. • Banking Account.
9 – 10	Formal Assessment	The weeks allocated for formal assessment are integrated across the weeks planned for teaching and learning. The assessment will consist of Practical Task/s with a 75% weighting and a Theory test with a 25% weighting.	
Activity 1- Practical Demonstration (Measuring, cutting, welding, grinding and drilling) = 25% of the Term mark. The Learners must be able to demonstrate certain skills acquired during the term. This is in the form of a single task and not part of the Terms practical model. A certain degree of difficulty needs to increase from term to term and the learner’s ability should be taken into consideration.			

Learner measures, cuts and prepares two pieces of material as per the drawing. The learner must produce tack welds that should be mechanically strong. A fillet weld should be performed at least 30mm, a hole needs to be drilled and accuracy is really important. All five stages must be assessed.

- Measuring
- Cutting
- Welding
- Drilling
- Grinding

Total for Activity 1 Marks to be converted to 25% of the total term mark.

Activity 2 – PRACTICAL ASSESSMENT TASK – PAT (Practical Project) = 50% of term mark.

Fundamental knowledge = 25%

Practical learning areas = 25%

Learners must now produce a practical project (PAT) using the all the method of construction (in Weeks allocated) which can be assessed to evaluate their acquisition skills. This project should involve the skills, techniques and knowledge of the theory component.

Assessments are done according to a rubric which includes all the skills acquired during the term. Marks total 50% of the term.

- Examples of projects are in the Addendum.

Activity 3 – Formal written or Oral Theory assessment = 25%. Scope of work that is done during the term.

Assessment: consists of two sections theory 25% and practical 75% = 100%

Annexure C: Theory Assessment: Oral and /or Written Pages

Theory section: (oral or written) score allocation 25%: comprises of all the above work done in weeks 1 through 10 inclusions should be pictures and multi answer questions. Pictures should constitute a major part of this section.

Annexure D: Practical Assessment: Pages

Practical section: Learners should now be ready to for an assessment with a use of hand tools so that the theoretical aspect can be implemented. Measurement now plays a role as this is the foundation to which all work can commence in the workshop.

At the end of this term; A project should be completed as evidence of the learner's ability.

(NB! Not all learners will be able to produce an exact reproduction of the example the educator has provided).

WEEK	TOPIC	CONTENT The learner is able to:	Techniques, activities, resources and process notes
1	Re-cap Safety and procedures	Demonstrate understanding of all safety and First Aid, HIV and AIDS procedures. Explain and demonstrate how to clean and dress a wound properly. Explain demonstrate the different procedures aligned with evacuating the workshop in an emergency situation.	Explain all safety, first aid and HIV and AIDS procedures, learners need to be shown all the signs and pictures to establish that safety should be first priority in the workshop. First aid procedures: • Educator to protect himself with gloves and apparel for HIV and Aids injury victims. • Examine the injury. • How to clean a wound. • Proper wound dressing • Reporting and written procedures when an injury has occurred. Emergency procedures: • Demonstrate a fire drill. • Emergency drill. • Power outage.

2	Communications and Graphics	Learner needs to understand and the use of the tools to engage in the proper design, layout and presentation of a drawing. • Pen, pencil, eraser, ruler and paper • Presentation and information • Correct scale and layout.	Proper introduction of graphics shows the necessary measurement, details that are required to do graphics. Introduce and explain the tools that are used for graphics • How to create a drawing. • Proper measurements. • Neatness and accuracy. • Proper layout.
3-4	Arc Welding process	Describe the shielded metal arc welding process. Weld carbon steel work-pieces using the shielded metal arc welding process Select, set up and conduct pre-operational checks of shielded metal arc welding equipment. Prepare work-pieces prior to welding. Weld work-pieces. Inspect welded work piece for defects in compliance with drawing specifications. Care for and store welding consumables and equipment.	The Educator must explain and demonstrate the importance of correct assembly of the shielded metal arc welding equipment: • The consequences of incorrect assembly, with reference to the vendor requirements. Identify and explain • Basic and major components in terms of manufacturer's requirements and standards. • Manufacturer's operational manual. • Worksite practices and safety and environmental issues. • Inspect work-pieces prior to welding.

			• Worksite procedures, tools, equipment, safety requirements, and materials.
5-6	Oxy-fuel gas cutting	Describe the oxy-fuel cutting process. Prepare for the oxy-fuel cutting operation. Cut material. Care and storage of cutting equipment, tools, and materials. Identify and explain basic and major components of the oxy-fuel cutting process and equipment, function and purpose is correct in terms of cutting standards. Care and storage of cutting equipment, tools, and materials	Explain and demonstrate the importance of correct setting of cutting pressures, and the consequences of incorrect settings. Cutting of material is carried out in accordance with work instruction sheet and drawing requirements. Inspect end product to conform to specifications as reflected on drawing or job requirement. Defects include excessive slag, rough cutting surface, jagged edges, rounded top corner. • Hazards include flashbacks: • Identify and correct cutting defects. • Material type to be used: May be selected from the range of carbon steels (plate only), applicable to the material groups 1, 2, 3 or 11 [according to ISO (TR) 15608]. • Material thickness: minimum - 10mm. • Positions: All positions/directions

			Explain the caring and storage procedures for tools, equipment in accordance with work site practices and specifications.
7	Welding process (Arc welder)	Describe the gas metal arc welding process and related equipment. Describe the Weld carbon steel work pieces using the gas metal arc welding process in the down-hand position. Describe the selection, assemble and conduct pre-operational checks of gas metal arc welding equipment. Describe the preparation work-pieces prior to welding and Weld work-pieces. Describe and inspect welded work-pieces for defects.	Explain the importance of correct assembly of the gas metal arc welding equipment, and the consequences of incorrect assembly, with reference to the manufacturer's requirements. Explain terms and use definitions consistent with generally accepted welding terminology as recorded in international welding standards; Suitable power source, wire-feeder, shielding gas, regulator, flow-meter, materials as specified on drawings and weld filler material, according to welding procedure specifications. Inspect work-pieces prior to welding. Resources include the following: - Worksite practices, tools and equipment. - Safety legislation as per job requirements. - Inspection methods - Visual, destructive or non-destructive.

		Describe the care for and store welding consumables and equipment.	Observations applicable prior to the tack-welding process: • Heat input. • Electrode size. • Filler-rod material selection. • Joint preparation. • Welding technique. • Consumable usage. • Handling and gas shielding. Demonstrate work-pieces welded in position. Demonstrate storage of welding consumables in accordance to worksite practices; tools and equipment stored to conform to worksite practices.
8	Entrepreneurship	Learners needs to understand the correct procedures, processes and hurdles when becoming and Entrepreneur. • Administration of a official documentation when opening a business. • Have the proper credit rating. • Funding to start a business.	Discuss and outline all the administration, legal and property selection in order to start a business. Important points need mentioning: • Type of business. • Partnership • Funding

		- Finding the proper property to start. - The overheads to succeed. - The staff to complement the type of business.	- Legal contracts. - Daily pitfalls in opening a business.
9 – 10	Formal Assessment	The weeks allocated for formal assessment are integrated across the weeks planned for teaching and learning. The assessment will consist of Practical Task/s with a 75% weighting and a Theory test with a 25% weighting.	
Activity 1- Practical Demonstration (Measuring, cutting, welding, grinding and drilling) = 25% of the Term mark. The Learners must be able to demonstrate certain skills acquired during the term. This is in the form of a single task and not part of the Terms practical model. A certain degree of difficulty needs to increase from term to term and the learner’s ability should be taken into consideration. Learner measures, cuts and prepares two pieces of material as per the drawing. The learner must produce tack welds that should be mechanically strong. A fillet weld should be performed at least 30mm, a hole needs to be drilled and accuracy is really important. All five stages must be assessed. - Measuring - Cutting - Welding - Drilling - Grinding Total for Activity 1 Marks to be converted to 25% of the total term mark. Activity 2 – PRACTICAL ASSESSMENT TASK – PAT (Practical Project) = 50% of term mark. Fundamental knowledge = 25%			

Practical learning areas = 25%

Learners must now produce a practical project (PAT) using the all the method of construction (in Weeks allocated) which can be assessed to evaluate their acquisition skills. This project should involve the skills, techniques and knowledge of the theory component.

Assessments are done according to a rubric which includes all the skills acquired during the term. Marks total 50% of the term.

- Examples of projects are in the Addendum.

Activity 3 – Formal written or Oral Theory assessment = 25%. Scope of work that is done during the term.

Assessment: consists of two sections theory 25% and practical 75% = 100%

Annexure C: Theory Assessment: Oral and /or Written Pages

Theory section: (oral or written) score allocation 25%: comprises of all the above work done in weeks 1 through 10 inclusions should be pictures and multi answer questions. Pictures should constitute a major part of this section.

Annexure D: Practical Assessment: Pages.....

Practical section: Learners should now be ready to for an assessment with a use of hand tools so that the theoretical aspect can be implemented. Measurement now plays a role as this is the foundation to which all work can commence in the workshop.

At the end of this term; A project should be completed as evidence of the learner's ability.

(NB! Not all learners will be able to produce an exact reproduction of the example the educator has provided).

WEEK	TOPIC	CONTENT The learner is able to:	Techniques, activities, resources and process notes
1	Re-cap Safety and procedures	Demonstrate understanding of all safety and First Aid, HIV and AIDS procedures. Explain and demonstrate how to clean and dress a wound properly. Explain and demonstrate the different procedures aligned with evacuating the workshop in an emergency situation.	Explain all safety, first aid and HIV and AIDS procedures, learners need to be shown all the signs and pictures to establish that safety should be first priority in the workshop. First aid procedures: • Educator to protect himself with gloves and apparel for HIV and Aids injury victims. • Examine the injury. • How to clean a wound. • Proper wound dressing • Reporting and written procedures when an injury has occurred. Emergency procedures: • Demonstrate a fire drill. • Emergency drill. • Power outage/ failure.

2	Re-cap on Graphics and Communications	Re-Cap Measure, estimate and calculate physical quantities and explore, describe and represent geometrical relationships in 2-dimensions in different life or workplace context	Correctly read scales on the measuring instruments and identify the appropriate instrument to measure a particular quantity. Carry out calculations correctly. Explain and demonstrate appropriate units measurement and calculation.
3-4	Core Welding Process	Describe the cored-wire welding process. Select, set up and conduct pre-operational checks of cored-wire welding equipment. Prepare work-pieces prior to welding and weld work-pieces. Inspect welded work-pieces for defects.	Educator must explain and demonstrate the importance of correct assembly of the cored-wire welding equipment, and the consequences of incorrect assembly, with reference to the vendor requirements. Educator must explain and demonstrate basic and major components of the cored-wire welding equipment in terms of manufacturer's requirements and standards. Educator must identify and explain parts and components used for the cored-wire welding process. Educator must explain terms and definitions used are consistent with generally accepted welding

		Care for and store welding consumables and equipment.	terminology as recorded in international welding standards. With explanation and demonstration of Welding equipment includes: Constant voltage power source, earth clamp, welding cable, shielding gas, flow meter regulator, hose clamps and welding consumables as specified. Demonstrate storage of welding consumables in accordance to worksite practices; tools and equipment stored to conform to worksite practices.
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5	Language and communication Financial and mathematics solutions	Explain sector identification and organisation type: • Sector/occupational focus: Services, manufacturing, financial, educational etc. • Organisation type: Government, parastatal, heavy/light industry, large organisation, small business Explain projected income and expenditure realistically. Demonstrate and define initial cost before work has started and profit that can be achieved when a product is completed.	Educator must describe and discuss features of occupational environment Description and discussion by the educator in which these features affect learning processes and/or application of learning Educator must carry out calculations using a calculator efficiently and correctly, and solutions obtained are verified in terms of the context. • Initial cost • Expenditure on products • Profit to be made.
6-7	Recap all the topics and work done for the year	Recap and reflecting, this will make sure that learners are properly prepared for Theory and Practical assessments:	Revise and reflecting, together with all relevant and related materials, wall charts, notes and power point presentations • Safety and first aid. • Tools. • Materials and joining. • Welding Equipment

			Recap and reflecting: together with all relevant and related materials, wall charts, notes and power point presentations. • Workshop Practices. • Material Selections. • Drawings and Shapes. Recap and reflecting: together with all relevant and related materials, wall charts, notes and power point presentations • Gas Arc Welding operations and processes • Oxy and acetylene cutting and welding • Cored-wire welding process. • Calculations • Language and communications • Mathematics and financial solutions
8	Re-Cap Entrepreneur	Learners understand the processes involved in becoming an Entrepreneur Requirements of becoming an Entrepreneur.	Revise the processes of becoming an Entrepreneur. There are Critical Stages You will go through as an Entrepreneur:

9 – 10	Formal Assessment	The weeks allocated for formal assessment are integrated across the weeks planned for teaching and learning. The assessment will consist of Practical Task/s with a 75% weighting and a Theory test with a 25% weighting.
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Activity 1- Practical Demonstration (Measuring, cutting, welding, grinding and drilling) = 25% of the Term mark.

The Learners must be able to demonstrate certain skills acquired during the term. This is in the form of a single task and not part of the Terms practical model. A certain degree of difficulty needs to increase from term to term and the learner's ability should be taken into consideration.

Learner measures, cuts and prepares two pieces of material as per the drawing. The learner must produce tack welds that should be mechanically strong. A fillet weld should be performed at least 30mm, a hole needs to be drilled and accuracy is really important. All five stages must be assessed.

- Measuring
- Cutting
- Welding
- Drilling
- Grinding

Total for Activity 1 Marks to be converted to 25% of the total term mark.

Activity 2 – PRACTICAL ASSESSMENT TASK – PAT (Practical Project) = 50% of term mark.

Fundamental knowledge = 25%

Practical learning areas = 25%

Learners must now produce a practical project (PAT) using the all the method of construction (in Weeks 1 – 10) which can be assessed to evaluate their acquisition skills. This project should involve the skills, techniques and knowledge of the theory component.

Assessments are done according to a rubric which includes all the skills acquired during the term. Marks total 50% of the term.

- Examples of projects are in the Addendum.

Activity 3 – Formal written or Oral Theory assessment = 25%. Scope of work that is done during the term.

Assessment: consists of two sections theory 25% and practical 75% = 100%

Annexure C: Theory Assessment: Oral and /or Written Pages

Theory section: (oral or written) score allocation 25%: comprises of all the above work done in weeks 1 through 10 inclusions should be pictures and multi answer questions. Pictures should constitute a major part of this section.

Annexure D: Practical Assessment: Pages

Practical section: Learners should now be ready to for an assessment with a use of hand tools so that the theoretical aspect can be implemented. Measurement now plays a role as this is the foundation to which all work can commence in the workshop.

At the end of this term; A project should be completed as evidence of the learner's ability.

(NB! Not all learners will be able to produce an exact reproduction of the example the educator has provided).

Year 4 Term 1

WEEK	TOPIC	CONTENT The learner is able to:	Techniques, activities, resources and process notes
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1	Introduction: (orientation) Safety & first aid	Demonstrate understanding of the following: • Safety procedures, • First boxes; • Protective clothes • Colours that signs are made out off and meaning. • Safety signs. • Sand buckets • Fire hoses. Apply how to clean and dress a wound Apply the following conditions: • Personal safety (PPE – Personal Protective Equipment) • Unsafe acts & conditions • Good house-keeping • General workshop rules • Demonstrate a workshop is cleaned and explain why. • What are unsafe conditions, the reasons for workshop to be kept safe and clean daily.	Theoretical explanation as well as practical demonstration. Videos and charts to be demonstrated. Worksheets: Oral and Written discussions: • In groups / teams. • Reference books and charts with suitable content • Practical activity. Role plays emergency procedures. • Practical activity. The use of sand buckets and fire hoses. Show how to clean and dress a wound properly and to register all injuries in a log book Worksheet and Video content on: • Safe workshop rules and practices. • First aid charts and videos show what happens in unsafe work conditions. • Introduce the learners to how the workshop is cleaned daily.
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		Learners need to apply the proper protocol for dealing with victims with HIV and AIDs. • Wearing gloves • Protective eye wear • Checking yourself for no open wounds • Treating the HIV and AIDs victim with caution • Proper disposal of all cleaning materials.	• The safe storage and care of all tools and equipment when learners are finish using the tools. Educator to demonstrate the HIV and Aids protocol: • Videos and charts • Oral discussions • Worksheets • Present information visually • Seek information from nursing sister.
2	Graphics & Communications	Measure, estimate and calculate physical quantities and explore, describe and represent geometrical relationships in 2-dimensions in different life or workplace context Estimate, measure and calculate physical quantities to solve problems in practical situations. Estimate, measure and calculate physical quantities to solve problems in practical situations.	Correctly read scales on the measuring instruments and identify the appropriate instrument to measure a particular quantity. Carry out calculations correctly. Explain and demonstrate appropriate unit measurement and calculation. Describe and identify properties of symmetrical shapes. Explain and identify concepts of lines of symmetry in 2-dimensional figures using paper folding and reflections in the lines of symmetry.

			Explain the concepts of transformation in terms of reflections, translations and rotations using concrete materials.
2	Communications & Graphics	A Learner will demonstrate the following: • Explain importance of graphics / drawing in the workshop. • Able to do rough / freehand drawings. • Scale up drawing to 1:1. • Do a material costing of a workshop project? Understand the importance of the proper way to cost a Job / project.	Educator: Introduction of Communications and Graphics. • Practical role of graphics. / drawings. • Explain the difference scale to drawings. • Demonstrate rough drawings. • Orthographic drawings. • Introduce learners to costing and profit making.
3	Tools	Explain the following: • Identify measuring tools and the proper use thereof. • Accurately measure length • Width. • Thickness in mm.	Educator with the aid of theory show learners hand tools and how to care and storage of them.

		Select use & care: Hand measuring tools. • Tape measure. • Steel ruler. • Meter stick. • Veneer Calliper. • Dividers callipers. • Combination square set. • Sliding adjustable square. • Height gauge with marble top.	Demonstrate the measurement equipment and why it is so important that these tools are taken care of and properly stored away. • Demonstrate how to measure and read in mm. • Educate how to use the measuring equipment. • Demonstrate how important accuracy is. • Explain how proper right angles are obtained. • Show learners' modern technology advancements. • How to mitre joints (45deg) properly.
4-5	Materials	Explain and demonstrate understanding of the following: • Ferrous & non-ferrous metals (profiles & use). • Show the physical difference between the two and the different applications where and why they are used. • Brass. • Bronze • Hardened Steel. • Copper	Practical demonstration by the educator for the following and the proper tests to be done to show the differentiation between non and ferrous metals. The following tests should be demonstrated and shown with the aid of wall charts and internet.

		• Aluminium. • Mild steel. • Stainless Steel. • Cast iron. • Galvanised Steel.	• The magnetic test. The use of a magnet and checking that the magnet sticks to the material. • The spark test. With the use of a grinding wheel the spark composition can be determined. • Show the different profiles that are used that materials are made in. • Explain and demonstrate the different types of materials • Profiles of the materials. • The different uses they have in the daily use in industry and in homes and why they needed to be used in all situations.

6-7	Welding equipment.	Describe the shielded metal arc welding process (SMAW). Explain and demonstrate the importance of correct assembly of the shielded metal arc welding equipment, and the consequences of incorrect assembly, with reference to the vendor requirements. Select, set up and conduct pre-operational checks of shielded metal arc welding equipment: Explain identify and connect an Arc weld for operation: • Suitable power source single and 3 phase • Earth clamp • Electrode holder • Welding settings. • Welding rod specifications for the appropriate job. • Maintenance of cables and helmets. • Proper storage of welding equipment and cables. • Proper ventilation when welding galvanised materials. Proper care of: • Welding helmet.	Educator should theoretically explain terms and definitions used with welding terminology as recorded in international welding standards. How to repair broken cables and to maintain the equipment. • Demonstrate the correct way to connect welding machine. • The correct polarity connections. • How to insert the welding rod properly in its holder. • The right equipment to use before you start to weld Carry out pre-operational checks in accordance with manufacturer's specifications. Resources to include: manufacturer's operational manual, worksite practices and safety. Demonstrate verify and selection shielded metal arc welding equipment as specified on welding procedure specification environmental issues.
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		• Replacement of welding helmet glasses (green & clear). • Safety glasses • Leather gloves. • Leather apron.	Educator must explain and demonstrate how to replace the broken welding helmet glasses (green and clear) also the proper way to adjust the head cradle adjustments. The correct amperage to be used to weld different gauge materials. The correct welding rods size to be used for different types of material. Identify and rectify hazards relating to welding process in accordance with standard work site practices. Educator must explain and demonstrate the correct procedures to be taken when welding galvanised materials. • Proper ventilation for the extraction of gases when welding galvanised materials. • Welding splatter when welding galvanised materials. • The correct preparation of materials before welding galvanised materials
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8	Entrepreneurship (Career Opportunities)	Learners understand the processes involved in becoming an Entrepreneur Requirements of becoming an Entrepreneur. Essential Skills Required to Succeed as an Entrepreneur • Ambition. It is easy to give up when the going gets tough, but the most successful entrepreneurs persist because of their ambitious nature. ... • Willingness to Learn. ... • Ability to Listen. ... • Creativity. ... • Assertiveness and Confidence. ... • Perseverance. ... • Courage and Risk Taking.	Introduce and explain the processes of becoming an Entrepreneur. There are Critical Stages You will go through as an Entrepreneur: • Finding The Right Business Idea: ... • Developing the business plan: ... • Raising Your Seed Funding: ... • Getting Paying Customers: ... • Success or Maybe Failure: Documentation for becoming an Entrepreneur (South African Citizen) • South African Citizen Identity Number. • South African Citizen Income Tax Number. • Proof of Residential Address. • Comprehensive Curriculum Vitae. • Banking Account.
9 – 10	Formal Assessment	The weeks allocated for formal assessment are integrated across the weeks planned for teaching and learning. The assessment will consist of Practical Task/s with a 75% weighting and a Theory test with a 25% weighting.	

Activity 1- Practical Demonstration (Measuring, cutting, welding, grinding and drilling) = 25% of the Term mark.

The Learners must be able to demonstrate certain skills acquired during the term. This is in the form of a single task and not part of the Terms practical model. A certain degree of difficulty needs to increase from term to term and the learner's ability should be taken into consideration.

Learner measures, cuts and prepares two pieces of material as per the drawing. The learner must produce tack welds that should be mechanically strong. A fillet weld should be performed at least 30mm, a hole needs to be drilled and accuracy is really important. All five stages must be assessed.

- Measuring
- Cutting
- Welding
- Drilling
- Grinding

Total for Activity 1 Marks to be converted to 25% of the total term mark.

Activity 2 – PRACTICAL ASSESSMENT TASK – PAT (Practical Project) = 50% of term mark.

Fundamental knowledge = 25%

Practical learning areas = 25%

Learners must now produce a practical project (PAT) using the all the method of construction (in Weeks 1 – 10) which can be assessed to evaluate their acquisition skills. This project should involve the skills, techniques and knowledge of the theory component.

Assessments are done according to a rubric which includes all the skills acquired during the term. Marks total 50% of the term.

- Examples of projects are in the Addendum.

Activity 3 – Formal written or Oral Theory assessment = 25%.

Scope of work that is done during the term.

Practical:

Activity 1: Fundamental section 25%. The learners should firstly select all safety equipment before starting assessment. Select and identify all the tools need to complete the project. Understand from the assessment worksheet what needs to be done to complete the assessment.

Activity 2: Practical Section 50%. Measure and mark the material to be used. Follow the Assessment worksheet to complete the project to be made.

Theory:

Activity 3: Theory section comprises of 25%. This is an oral/ written section whichever the Learner is comfortable with. The answer block section helps the learners to choose the correct answer as well as the correct identification of colour pictures of related equipment

WEEK	TOPIC	CONTENT The learner is able to:	Techniques, activities, resources and process notes
1	Re-cap Safety and procedures	Demonstrate understanding of the following • Safety procedures, • First Aid boxes and the positions of such boxes. • Protective clothes and their safety features. • Colours that signs are made out off and meaning. • Safety signs. Explain how to clean and dress a wound Apply the following conditions: • Personal safety (PPE – Personal Protective Equipment). • Unsafe acts & conditions. • Good house-keeping. • General workshop rules and practices. • Demonstrate a workshop is cleaned and explain why. • What unsafe conditions are, the reasons for workshop to be kept safe and clean daily.	Theoretical explanation as well as practical demonstration. Videos and charts to be demonstrated. Worksheets: Oral and Written discussions: • In groups / teams while educator gives the proper explanations. • Reference books and charts with suitable content. • Practical activity. Role plays emergency procedures. • The correct colour coding of all relative signs that are used in the workshop. Show how to clean and dress a wound properly and to register all injuries in a log book. Worksheet and Video content on: • Safe workshop rules and practices.

		• Safe evacuation procedures in the case of fire and emergencies. Explain the proper protocol for dealing with victims with HIV and AIDs. • Wearing gloves • Protective eye wear • Checking yourself for no open wounds • Treating the HIV and AIDs victim with caution • Proper disposal of all cleaning materials.	• First aid charts and videos show what happens in unsafe work conditions. • Introduce the learners to how the workshop is cleaned daily. • Safe evacuation of the workshop. • The correct fire drill and evacuation procedure when a fire and/or emergency happens. Educator to demonstrate and discuss the HIV and Aids protocol: • Videos and charts • Oral discussions • Worksheets • Present information visually • Seek information from nursing sister.
2	Communications and Graphics	Learner needs to understand and the use of the tools to engage in the proper design, layout and presentation of a drawing. • Pen, pencil, eraser, ruler and paper • Presentation and information	Proper introduction of graphics shows the necessary measurement, details that are required to do graphics. Introduce and explain the tools that are used for graphics • How to create a drawing.

		Correct scale and layout.	• Proper measurements. • Neatness and accuracy. Proper layout.
3-4	Apply work site practices.	Explain the selection, use and care for engineering hand tools and hand power tools. Hand tools examples: • Dividers • G clamps • Cole chisel • Various files and uses • Scroll benders • Ball peen hammer • Anvil • Spanners and sockets Power hand tools examples: • Compressor. • Pneumatic hand tools. • Hand drill. • Angle grinders 250/350mm. • Finishing grinder. • Jig saw.	Theoretical explanation and demonstrations of the topics: • The theory can be explained while doing a practical demonstration but pen on paper is important. • Provide the learner with the appropriate note where textbooks are not available. • Attention must be given to correct terminology, safe use and proper care of tools is required. • All topics and content should be covered in theory. Demonstrate the correct way in which power tools need to be stored and also the way electrical cabling needs to be properly rolled up.

		• SDS rotary drill. Floor mounted equipment: • Folder • Bender • Bench grinder • Guillotine. • Pedestal drill. • 300mm diameter disc sander. Explain how the following: • Repair and replace a faulty 3 core cable. • Repair and replace a faulty 3 pin plug. • Store and clean hand power tools.	Educator needs to explain demonstrate the following: • Repairs to power cables. • Replacing power cables • Replacing the plug top • Replacing brushes and bearings inside power hand tools and motors. • Proper storage and care of all tools located in the lockable area.
5-6	Material selection	Explain, identify and select material to specification. Plan the identification process to identify materials to specification on. explain and demonstrate the steps in the identification of materials to specification to be in logical sequence and to job requirements. The correct application for the following materials:	Demonstrate all safety precautions in relation to work site practices and procedures. Educator needs to explain and demonstrate the following tests with the aid of wall charts and internet. • The magnetic test. The use of a magnet and checking that the magnet sticks to the material.

		• Galvanised materials. • Mild steel. • Stainless Steel. • Aluminium. • Carbon steel. • Brass and copper. Identify materials in accordance with material specification sheet as included in work instructions.	• The spark test. With the use of a grinding wheel the spark composition can be determined. Identify materials marked with colour code or identification numbers. • Materials to be identified are limited to: • Mild steel. • Galvanised steel. • Aluminium. • Stainless steel. • Carbon steels. • Brass and copper. Identification methods visually with the aid of de-coding systems and mechanical processes
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7	Welding	Learner discuss and demonstrate the different types of Non-Ferrous and Ferrous Metals and the following: • Select and identify materials. • How to do a magnetic test on them. • A Spark test. • Oxidation processes (rust and galvanising) • What type of joining is associated with the materials.	Display samples of Ferrous and Non Ferrous, discussing the properties, how to do a spark and magnetic test. Displaying and visual stimulus: • Video and chart displays. • Cut-off materials for selection. • Magnets for testing. • Bench grinder (Spark test). • Rust and galvanising. • Joining methods.
8	Entrepreneurship	Learners need to understand the correct procedures, processes and hurdles when becoming an Entrepreneur. • Administration of an official documentation when opening a business. • Have the proper credit rating. • Funding to start a business. • Finding the proper property to start. • The overheads to succeed.	Discuss and outline all the administration, legal and property selection in order to start a business. Important points need mentioning: • Type of business. • Partnership • Funding • Legal contracts. Daily pitfalls in opening a business.

		The staff to complement the type of business.	
9 – 10	Formal Assessment	The weeks allocated for formal assessment are integrated across the weeks planned for teaching and learning. The assessment will consist of Practical Task/s with a 75% weighting and a Theory test with a 25% weighting. Activity 1- Practical Demonstration (Measuring, cutting, welding, grinding and drilling) = 25% of the Term mark. The Learners must be able to demonstrate certain skills acquired during the term. This is in the form of a single task and not part of the Terms practical model. A certain degree of difficulty needs to increase from term to term and the learner's ability should be taken into consideration. Learner measures, cuts and prepares two pieces of material as per the drawing. The learner must produce tack welds that should be mechanically strong. A fillet weld should be performed at least 30mm, a hole needs to be drilled and accuracy is really important. All five stages must be assessed. • Measuring • Cutting	

- Welding
- Drilling
- Grinding

Total for Activity 1 Marks to be converted to 25% of the total term mark.

Activity 2 – PRACTICAL ASSESSMENT TASK – PAT (Practical Project) = 50% of term mark.

Fundamental knowledge = 25%

Practical learning areas = 25%

Learners must now produce a practical project (PAT) using the all the method of construction (in Weeks 1 – 10) which can be assessed to evaluate their acquisition skills. This project should involve the skills, techniques and knowledge of the theory component.

Assessments are done according to a rubric which includes all the skills acquired during the term. Marks total 50% of the term.

- Examples of projects are in the Addendum.

Activity 3 – Formal written or Oral Theory assessment = 25%.

Scope of work that is done during the term.

Practical:

Activity 1: Fundamental section 25%. The learners should firstly select all safety equipment before starting assessment. Select and identify all the tools need to complete the project. Understand from the assessment worksheet what needs to be done to complete the assessment.

Activity 2: Practical Section 50%. Measure and mark the material to be used. Follow the Assessment worksheet to complete the project to be made.

Theory:

Activity 3: Theory section comprises of 25%. This is an oral/ written section whichever the Learner is comfortable with. The answer block section helps the learners to choose the correct answer as well as the correct identification of colour pictures of related equipment.

WEEK	TOPIC	CONTENT The learner is able to:	Techniques, activities, resources and process notes
1	Re-cap Safety and procedures	Demonstrate understanding of all safety and First Aid, HIV and AIDS procedures. Explain and demonstrate how to clean and dress a wound properly. Explain and demonstrate the different procedures aligned with evacuating the workshop in an emergency situation.	Explain all safety, first aid and HIV and AIDS procedures, learners need to be shown all the signs and pictures to establish that safety should be first priority in the workshop. First aid procedures: • Educator to protect himself with gloves and apparel for HIV and Aids injury victims. • Examine the injury. • How to clean a wound. • Proper wound dressing • Reporting and written procedures when an injury has occurred. Emergency procedures: • Demonstrate a fire drill • Emergency drill • Power outage

2	Communications and Graphics	Measure, estimate and calculate physical quantities and explore, describe and represent geometrical relationships in 2-dimensions in different life or workplace context • Estimate • Measure • Calculate physical quantities. • To solve problems in theoretical situations. To solve problems in practical I situations.	Correctly read scales on the measuring instruments and identify the appropriate instrument to measure a particular quantity. Carry out calculations correctly. Explain and demonstrate appropriate unit measurement and calculation. Describe and identify properties of symmetrical shapes. Explain and identify concepts of lines of symmetry in 2-dimensional figures using paper folding and reflections in the lines of symmetry. Explain the concepts of transformation in terms of reflections, translations and rotations using concrete materials.
3-4	Arc Welding process	Description by learners of the shielded metal arc welding process (SMAW). Weld carbon steel work-pieces using the shielded metal arc welding process Select, set up and conduct pre-operational checks of shielded metal arc welding equipment.	Explain and demonstrate the importance of correct assembly of the shielded metal arc welding equipment, and the consequences of incorrect assembly, with reference to the vendor requirements.

		Prepare work-pieces prior to welding. Weld work-pieces. Inspect welded work piece for defects in compliance with drawing specifications. Care for and store welding consumables and equipment.	Identify and explain basic and major components of the shielded metal arc welding equipment in terms of manufacturer's requirements and standards. Resources to include: • Manufacturer's operational manual, • Worksite practices and safety and environmental issues Inspect work-pieces prior to welding. • Worksite procedures, tools, equipment, safety requirements, and materials.
5-6	Oxy-fuel gas cutting	Describe the oxy-fuel cutting process. Prepare for the oxy-fuel cutting operation. • Cut material. • Care and storage of cutting equipment, tools, and materials. Identify and explain basic and major components of the oxy-fuel cutting process and equipment, function and purpose is correct in terms of cutting standards.	The educator must explain and demonstrate the following: • Importance of correct setting of cutting pressures, and the consequences of incorrect settings. • Cutting of material is carried out in accordance with work instruction sheet and drawing requirements. • Inspect end product to conform to specifications as reflected on drawing or job requirement.

		Care and storage of cutting equipment, tools, and materials	• Defects include excessive slag, • Rough cutting surface, • Jagged edges, rounded top corner. • Hazards include flashbacks: • Identify and correct cutting defects. Material type to be used: May be selected from the range of carbon steels (plate only), applicable to the material groups 1, 2, 3 or 11 [according to ISO (TR) 15608]. • Material thickness: minimum - 10mm. • Positions: All positions/directions • Explain the caring and storage procedures for tools, equipment in accordance with work site practices and specifications.
7	Welding process (Arc welder)	Describe the gas metal arc welding process and related equipment. Weld carbon steel work pieces using the gas metal arc welding process in the down-hand position Select, assemble and conduct pre-operational checks of gas metal arc welding equipment.	Explanation by the educator demonstrating the importance of correct assembly of the gas metal arc welding equipment: • The consequences of incorrect assembly, with reference to the manufacturer's requirements.

		Prepare work-pieces prior to welding and Weld work-pieces. Inspect welded work-pieces for defects. Explain the process of how to inspect welded work-pieces for defects. • Porosity. • In correct welding flow/run. • Brittle joints. • Under cutting. Explain the process of how to care for and store welding consumables and equipment.	Explain terms and use definitions consistent with generally accepted welding terminology as recorded in international welding standards: • Suitable power source. • Wire-feeder. • Shielding gas, • Regulator, • Flow-meter, • Materials as specified on drawings. • Weld filler material, according to welding procedure specifications. Inspect work-pieces prior to welding. Resources include for the educator's explanation and demonstration of the following: • Worksite practices, • Tools and equipment. • Safety legislation as per job requirements. • Inspection methods - Visual, destructive or non-destructive. Observations applicable prior to the tack-welding process: Heat input, electrode size, filler-rod material selection, joint preparation, welding
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			technique, consumable usage, handling and gas shielding. Demonstrate work-pieces welded in position. Demonstrate storage of welding consumables in accordance to worksite practices; tools and equipment stored to conform to worksite practices.
8	Entrepreneurship	Learners needs to understand the correct procedures, processes and hurdles when becoming and Entrepreneur. • Administration of an official documentation when opening a business. • Have the proper credit rating. • Funding to start a business. • Finding the proper property to start. • The overheads to succeed. The staff to complement the type of business.	Discuss and outline all the administration, legal and property selection in order to start a business. Important points need mentioning: • Type of business. • Partnership • Funding • Legal contracts. Daily pitfalls in opening a business.

9 – 10	Formal Assessment	The weeks allocated for formal assessment are integrated across the weeks planned for teaching and learning. The assessment will consist of Practical Task/s with a 75% weighting and a Theory test with a 25% weighting.
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Activity 1- Practical Demonstration (Measuring, cutting, welding, grinding and drilling) = 25% of the Term mark.

The Learners must be able to demonstrate certain skills acquired during the term. This is in the form of a single task and not part of the Terms practical model. A certain degree of difficulty needs to increase from term to term and the learner's ability should be taken into consideration.

Learner measures, cuts and prepares two pieces of material as per the drawing. The learner must produce tack welds that should be mechanically strong. A fillet weld should be performed at least 30mm, a hole needs to be drilled and accuracy is really important. All five stages must be assessed.

- Measuring
- Cutting
- Welding
- Drilling
- Grinding

Total for Activity 1 Marks to be converted to 25% of the total term mark.

Activity 2 – PRACTICAL ASSESSMENT TASK – PAT (Practical Project) = 50% of term mark.

Fundamental knowledge = 25%

Practical learning areas = 25%

Learners must now produce a practical project (PAT) using the all the method of construction (in Weeks allocated) which can be assessed to evaluate their acquisition skills. This project should involve the skills, techniques and knowledge of the theory component.

Assessments are done according to a rubric which includes all the skills acquired during the term. Marks total 50% of the term.

- Examples of projects are in the Addendum.

Activity 3 – Formal written or Oral Theory assessment = 25%.

Scope of work that is done during the term.

Practical:

Activity 1: Fundamental section 25%. The learners should firstly select all safety equipment before starting assessment. Select and identify all the tools need to complete the project. Understand from the assessment worksheet what needs to be done to complete the assessment.

Activity 2: Practical Section 50%. Measure and mark the material to be used. Follow the Assessment worksheet to complete the project to be made.

Theory:

Activity 3: Theory section comprises of 25%. This is an oral/ written section whichever the Learner is comfortable with. The answer block section helps the learners to choose the correct answer as well as the correct identification of colour pictures of related equipment.

WEEK	TOPIC	CONTENT	Techniques, activities, resources and process notes
		Revision and consolidation	
1	Re-cap Safety and procedures	Insure that learners are properly updated with all safety and First Aid, HIV and AIDS procedures. Demonstrate how to clean and dress a wound properly. Demonstrate the different procedures aligned with evacuating the workshop in an emergency situation.	Explain all safety, first aid and HIV and AIDS procedures, learners need to be shown all the signs and pictures to establish that safety should be first priority in the workshop. First aid procedures: • Educator to protect himself with gloves and apparel for HIV and Aids injury victims. • Examine the injury. • How to clean a wound. • Proper wound dressing • Reporting and written procedures when an injury has occurred. Emergency procedures: • Demonstrate a fire drill. • Emergency drill. • Power outage/ failure.

	Re-cap on Graphics and Communications	Re-Cap Measure, estimate and calculate physical quantities and explore, describe and represent geometrical relationships in 2-dimensions in different life or workplace context	Correctly read scales on the measuring instruments and identify the appropriate instrument to measure a particular quantity. Carry out calculations correctly. Explain and demonstrate appropriate unit measurement and calculation.
2	Core Welding Process	Describe the cored-wire welding process. Select, set up and conduct pre-operational checks of cored-wire welding equipment. Prepare work-pieces prior to welding and weld work-pieces. Inspect welded work-pieces for defects.	Educator needs to explain and demonstrate the importance of correct assembly of the cored-wire welding equipment: • The consequences of incorrect assembly. • Reference to the vendor requirements. • Explain and demonstrate basic and major components. • The manufacturer's requirements and standards. • Identify and explain parts and components. • The definitions used are consistent with generally accepted welding terminology as recorded in international welding standards. Educator needs to explain and demonstrate the Welding_equipment includes:

		Explain the process of how to care for and store welding consumables and equipment.	• Constant voltage power source. • Earth clamp and welding cables. • Shielding gas. • Flow meter regulator. • Hose clamps and welding consumables as specified. Demonstrate storage of welding consumables in accordance to worksite practices; tools and equipment stored to conform to worksite practices.
3	Language and communication Financial and mathematics solutions	Sector identification and organisation type: • Sector/occupational focus: Services, manufacturing, financial, educational etc. Organisation type: Government, parastatal, heavy/light industry, large organisation, small business Explain projected income and expenditure realistically. Demonstrate and define initial cost before work has started and profit that can be achieved when a product is completed.	Educator needs to describe and discuss features of occupational environment Described and discuss ways in which these features affect learning processes and/or application of learning Educator needs to carry out calculations using a calculator efficiently and correctly, and solutions obtained are verified in terms of the context. • Initial cost • Expenditure on products

			• Profit to be made.
4	Re-Capping, revision of the terms work in relation to Entrepreneurship	Revision of the terms work: Learners understand the processes involved in becoming an Entrepreneur Requirements of becoming an Entrepreneur. • Administration • Funding • Pit-falls • Resolutions • Labour Laws.	Revise the processes of an Entrepreneurship. This will entail all the aspects of the following: • Administration • Funding • Pit-falls • Resolutions • Labour Laws. There are Critical Stages You will go through as an Entrepreneur:

5-10	External examination	External moderation of school assessment over terms 1, 2 and 3 = 75% of qualification Formal external assessment written test or oral = 25% of qualification	

SECTION 4:

ASSESSMENT IN WELDING

4.1 INTRODUCTION

Assessment is a continuous, planned process of identifying, gathering and interpreting information about the performance of learners, using various forms of assessment. It involves four steps: generating and collecting evidence of achievement; evaluating this evidence; recording the findings and using this information to understand and thereby assist the learner's development in order to improve the process of learning and teaching.

Assessment should be both informal (Assessment for Learning) and formal (Assessment of Learning). In both cases regular feedback should be provided to learners to enhance the learning experience.

4.2 INFORMAL OR DAILY ASSESSMENT

Assessment for learning has the purpose of continuously collecting information about learner performance, that can be used to improve their learning.

Informal assessment is a daily monitoring of learners' progress. This is done through observations, discussions, practical demonstrations, learner-teacher conferences, informal classroom interactions, etc. Informal assessment may be as simple as stopping during the lesson to observe learners or to discuss with learners how learning is progressing. Informal assessment should be used to provide feedback to learners and to inform planning for teaching, but need not be recorded. It should not be seen as separate from the learning activities taking place in the classroom.

Self-assessment and peer assessment actively allow learners to assess themselves. This is important as it allows learners to learn from, and reflect on their own performance. The results of the informal daily assessment tasks are not formally recorded unless the teacher wishes to do so. The results of daily assessment tasks are not taken into account for promotion purposes.

4.3 FORMAL ASSESSMENT

All assessment tasks that make up a formal programme of assessment for the year are regarded as Formal Assessment. Formal assessment tasks are marked and formally recorded by the teacher for progression and certification purposes. All Formal Assessment tasks are subject to moderation for the purpose of quality assurance and to ensure that appropriate standards are maintained.

Formal assessment provides teachers with a systematic way of evaluating how well learners are progressing in a Year and in a particular subject. Examples of formal assessments include tests, examinations, practical tasks, projects, oral presentations, demonstrations, performances, etc.

Formal assessment tasks form part of a yearlong formal Programme of Assessment in each Year and subject, are school-based and are weighted as follows for the different grades:

In **Year one** a learner is exposed to a number of Occupational Subjects. Each subject is offered over a ten-week period (one term) in Year 1, where the learner is exposed to the basic skills required for the subject. By the end of Year 1 the learner will select a minimum of one skill for the qualification.

Year 1	Formal School-Based Assessments
	Learner performance in the Term:
	Practical 75%
	Theory 25%
Term Report	100%

Years 2 and 3

Year 2 will focus on a broad overview of the subject with a basic understanding and mastery of some of the basic skills required in the subject. Year 3 will focus on the consolidation of the basic skills and the addition of more advanced skills. Learners must in Year 3 start to develop a greater degree of independent mastery of the subject skills.

Year 2/3	Formal School-Based Assessments			Final End-of-Year Assessments
	Term 1	Term 2	Term 3	Term 4
	Practical 75%	Practical 75%	Practical 75%	○ Practical 75%
	Theory 25%	Theory 25%	Theory 25%	○ Pen and Paper Test/ Exam 25%
Term Report	100%	100%	100%	
End of Year	CASS 75%			25%

Year 4 Qualification year

In year 4 the focus shifts to the World of Work. Learners must consolidate required skills for the qualification and may engage in workplace exposure for a short period of time during the fourth year. Learners develop independent mastery of skills to be competent within the workplace

Year 4	Formal School-Based Assessments			Final End-of-Year Assessments
	Term 1	Term 2	Term 3	Term 4
	Practical 75%	Practical 75%	Practical 75%	External Practical Assessment Task 25%
	Theory 25%	Theory 25%	Theory 25%	
Term Report	100%	100%	100%	External Test/ Exam 25%
End of Year	School Based Assessment (SBA) = 50%			External Assessments = 50%
Promotion/ Progression	SBA + EA = Promotion/Progression 50% + 50% = 100%			

CLARIFICATION ON ASSESSMENT PERIODS

Year 2 and 3:

Term 1 theory assessment to consist of work done in term 1 only

Term 2 theory assessment to consist of work done in terms 1 and 2

Term 3 theory assessment to consist of work done in term 3 only

Term 4 theory assessment to consist of work done in terms 3 and 4

Year 4:

Term 1 theory assessment to consist of work done in term 1 only

Term 2 theory assessment to consist of work done in terms 1 and 2

Term 3 theory assessment to consist of work done in terms 1, 2 and 3

Term 4 Theory completed in the year

The cognitive demands in assessment should be **appropriate for the age and developmental level** of the learners in the grade. Assessment in Bricklaying and Plastering must cater for a range of cognitive levels and abilities of learners. The assessment tasks should be carefully designed to cover the content of the subject as well as the range of skills and the cognitive levels that have been identified in the specific aims. The design of assessments should therefore ensure that a full range

of content and skills are assessed within each Year. The specific aims, topics, content and range of skills in the subject should be used to inform the planning and development of assessments. Formal assessments must cater for a range of cognitive levels and abilities of learners, as shown below:

SUGGESTED WEIGHTING FOR COGNITIVE LEVELS: THEORY			
Cognitive level	Year 1	Year 2 - 3	Year 4
Lower order: Remembering	70%	60%	50%
Middle order: Understanding Applying	20%	30%	40%
Higher order: Analysing / Evaluating and Creating	10%	10%	10%

SUGGESTED WEIGHTING FOR COGNITIVE LEVELS: PRACTICAL WORK			
Cognitive level	Year 1	Year 2 - 3	Year 4
Lower order: Isolated single steps	70%	60%	50%
Middle order: Multiple limited steps	20%	30%	40%
Higher order: Several operations required, Creative production/design	10%	10%	10%

Examiners and moderators **in all subjects** are encouraged to think about difficulty of each question. Levels of difficulty are categorised as follows:

- D1: Easy for the envisaged Year 1 - 4 candidate to answer
- D2: Moderately challenging for the envisaged Year 1 - 4 candidate to answer
- D3: Difficult for the envisaged Year 1 - 4 candidate to answer

In judging the level of difficulty of each question, both the demands that each question makes on the cognitive ability of an average Year 1 - 4 learner and the intrinsic difficulty of the question or task

is considered. In making this judgement, the difficulty or ease of a particular question is identified. A four-category framework for thinking about question or item difficulty adapted from Leong (2006) has been used in this identification process. This framework comprises the following four general sources of difficulty:

- **Content difficulty:** this indexes the difficulty of the subject matter, topic or conceptual knowledge; some content is inherently more difficult than other content.

- **Stimulus difficulty:** this relates to the linguistic features of the question and the challenge that candidates face in reading, interpreting and understanding the question.
- **Task difficulty:** this refers to the difficulty that candidates face when trying to formulate or produce an answer.
- **Expected response difficulty:** this refers to difficulties because of the mark scheme or memorandum, in other words how marks are allocated.

Examiners and moderators **in all subjects** are encouraged to think about difficulty of each question. Levels of difficulty are categorised as follows:

D1: Easy for the envisaged Year 1 - 4 candidate to answer

D2: Moderately challenging for the envisaged Year 1 - 4 candidate to answer

D3: Difficult for the envisaged Year 1 - 4 candidate to answer

In judging the level of difficulty of each question, both the demands that each question makes on the cognitive ability of an average Year 1 - 4 learner and the intrinsic difficulty of the question or task

is considered. In making this judgement, the difficulty or ease of a particular question is identified. A four-category framework for thinking about question or item difficulty adapted from Leong (2006) has been used in this identification process. This framework comprises the following four general sources of difficulty:

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- **Stimulus difficulty:** this relates to the linguistic features of the question and the challenge that candidates face in reading, interpreting and understanding the question.
- **Task difficulty:** this refers to the difficulty that candidates face when trying to formulate or produce an answer.
- **Expected response difficulty:** this refers to difficulties because of the mark scheme or memorandum, in other words how marks are allocated.

4.4 Suggested Program of Assessment for Welding

YEAR 1					
Term	Content/ concept/skill	Activities	Forms of Assessment	%	FATs based on activities in CAPS: SV

Year 1	• Health & Safety • Workshop Layout • Equipment • Measurement • Hand tools • Drawings • Materials • Methods of joining • Opportunities in the industry	Activity 1 Exercise in drawings and measurement	Practical Demonstration	25%	FAT 1
		Activity 2 Model	Practical	50%	
		Activity 3 Respond to questions	Pen and paper test (Oral or written)	25%	

YEAR 2					
Term	Content/ concept/skill	Activities	Forms of Assessment	%	FATs based on activities in CAPS: SV
Term 1	• Health & Safety • Workshop Layout • Equipment • Measurement • Hand tools • Drawings • Materials • Methods of joining	Activity 1 Measuring, Cutting & Tack weld	Demonstration	25%	FAT 1
		Activity 2 Model (PAT)	Practical	50%	
		Activity 3 Respond to questions	Pen and paper test (Oral or written)	25%	
Term 2	• Safety • Hand tools	Activity1 Model piece demonstrating a	Demonstration	15%	FAT 2

	- Power Tools - Materials - Methods of joining	tack and fillet weld.			
		Activity 2 Model	Practical	60%	
		Activity 3 Respond to questions	Pen and paper test (Oral or written)	25%	
Term 3	- Safety - Hand tools - Power Tools - Drawings - Materials - Welding process- Arc welding equipment	Activity 1 Demonstration	Demonstration	35%	FAT 3
		Activity 2 Model	Demonstration	40%	
		Activity 3 Respond to questions	Theory: Oral/ Written	25%	

Term 4	- Safety - Welding operations - Gas Arc Welding - Calculations	Activity 1 Model (PAT)	Practical	75%	FAT 4
		Activity 2 Respond to questions	Pen and paper test (Oral or written)	25%	

YEAR 3					
Term	Content/ concept/skill	Activities	Forms of Assessment	%	FATs based on

					activities in CAPS: SV
Term 1	- Health & Safety - Equipment - Measurement - Hand tools - Materials - Welding Process-Arc Welder equipment. - Calculations	Activity1 Demonstration	Demonstration	25%	FAT 1
		Activity2 Model	Practical	50%	
		Activity3 Respond to questions	Pen and paper test (Oral or written)	25%	
Term 2	- Safety - Equipment - Hand tools - Power Tools - Materials Selection - Measuring - Drawings & Shapes	Activity1 Demonstration	Demonstration	15%	FAT 2
		Activity2 Model	Practical	60%	
		Activity3 Respond to questions	Pen and paper test (Oral or written)	25%	
Term 3	- Safety - Arc welding Process - Oxy Fuel cutting - Welding-Arc welder - Core Welding Process	Activity 1 Demonstration	Demonstration	25%	FAT 3
		Activity 2 Model	Demonstration	50%	
		Activity 3 Respond to questions	Pen and paper test (Oral or written)	25%	
Term 4	Safety	Activity 1 Model	Practical	75%	FAT 4

	- Core Welding process - Tools and equipment - Materials - Joining methods	Activity 2 Respond to questions	Pen and paper test (Oral or written)	25%	
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YEAR 4					
Term	Content/ concept/skill	Activities	Forms of Assessment	%	FATs based on activities in CAPS: SV
Term 1	- Safety and first aid - Tools - Equipment - Materials - Measurements - Welding process - Joining methods	Activity 1 Demonstration	Demonstration	25%	FAT 1
		Activity 2 Model	Practical	50%	
		Activity 3 Respond to questions	Pen and paper test (Oral or written)	25%	
Term 2	- Safety and first aid - Work site practices - Materials - Material selection - Drawings and shapes	Activity 1 Model	Practical	75%	FAT 2
		Activity 2 Respond to questions	Pen and paper test (Oral or written)	25%	
Term 3	- Safety and first aid - Arc welding process	Activity 1 Demonstration	Demonstration	25%	FAT 3

	• Oxy fuel cutting • Welding process (Arc) • Joining methods	Activity 2 Model	Practical	50%	
		Activity 3 Respond to questions	Pen and paper test (Oral or written)	25%	
Term 4	Core content and Concept across the year	External moderation of school assessment over terms 1, 2 and 3.		75%	
		Activity 1 Respond to questions	Formal external assessment: Written test (or oral where necessary)	25%	

4.5 RECORDING AND REPORTING

Recording is a process in which the teacher documents the level of a learner's performance in a specific assessment task. It indicates learner progress towards the achievement of the knowledge as prescribed in the Curriculum and Assessment Policy Statements. Records of learner performance should provide evidence of the learner's conceptual progression within a Year and her/his readiness to progress or promote to the subsequent Year. Records of learner performance should also be used to verify the progress made by teachers and learners in the teaching and learning process.

Reporting is a process of communicating learner performance to learners, parents, schools, and other stakeholders. Learner performance can be reported in a number of ways. These include report cards, parents' meetings, school visitation days, parent-teacher conferences, phone calls, letters, class or school newsletters, etc. Teachers in all grades report in percentages against the subject. Seven levels of competence have been described for each subject listed for Grades R - 12. The various achievement levels and their corresponding percentage bands are as shown in the Table below.

Codes and Percentages for Recording and Reporting

Rating code	Description of competence	Percentage
7	Outstanding achievement	80 - 100
6	Meritorious achievement	70 - 79

5	Substantial achievement	60 - 69
4	Adequate achievement	50 - 59
3	Moderate achievement	40 - 49
2	Elementary achievement	30 - 39
1	Not achieved	0 - 29

Note: The seven-point scale should have clear descriptors that give detailed information for each level.

Teachers will record actual marks against the task by using a record sheet; and report percentages against the subject on the learners' report cards.

4.6 MODERATION OF ASSESSMENT

Moderation refers to the process that ensures that the assessment tasks are fair, valid and reliable.

Moderation should be implemented at school, district, provincial and national levels.

Comprehensive and appropriate moderation practices must be in place for the quality assurance of all subject assessments.

Formal assessment (SBA)

Moderation at the school will be carried out at least once each school term by the Head of Department responsible for the subject. This moderation needs to take place before cluster, district, regional or provincial moderation. Teachers' portfolios and evidence of learner performance must be moderated to ensure that a variety of assessment tasks have been used to address various topics and that assessment covered a range of cognitive levels.

4.7 GENERAL

This document should be read in conjunction with:

4.7.1 *National policy pertaining to the programme and promotion requirements of the National Curriculum Statement Grades R-12; and*

4.7.2 *National Protocol for Assessment Grades R-12;*

4.7.3 *White Paper 6 on Special Needs Education: Building an Inclusive Education and Training System (2001);*

4.7.4 *Guidelines for Responding to Diversity in the Classroom through the Curriculum and Assessment Policy Statements (2011);*

4.7.5 *Guidelines to Ensure Quality Education and Support in Special Schools and Special School Resource Centres (2013);*

4.7.6 *Policy on Screening, Identification, Assessment and Support (2014);*

4.7.7 *Guidelines for Full-service/Inclusive Schools (2010); and*

4.7.8 *Standard Operating Procedures for Assessment of Learners who Experience Barriers to Assessment (2016).*